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(19) **United States**(12) **Patent Application Publication**
CARLO, III et al.(10) **Pub. No.: US 2025/0275774 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **ORTHOPEDIC STAPLE AND RELATED INSTRUMENTS****Publication Classification**(71) Applicant: **WRIGHT MEDICAL TECHNOLOGY, INC.**, Memphis, TN (US)(72) Inventors: **Robert Michael CARLO, III**, Lakeland, TN (US); **Zachary KORMAN**, St. Louis, MO (US); **Shannon D. CUMMINGS**, Hernando, MS (US); **John Kent ELLINGTON**, Charlotte, NC (US); **John S. LEWIS, Jr.**, Prospect, KY (US)(73) Assignee: **WRIGHT MEDICAL TECHNOLOGY, INC.**, Memphis, TN (US)(51) **Int. Cl.**

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A61B 17/56 (2006.01)
A61F 2/08 (2006.01)

(52) **U.S. Cl.**

CPC *A61B 17/17* (2013.01); *A61B 2017/00668* (2013.01); *A61B 17/064* (2013.01); *A61B 2017/0641* (2013.01); *A61B 17/0642* (2013.01); *A61B 2017/0646* (2013.01); *A61B 17/068* (2013.01); *A61B 17/1682* (2013.01); *A61B 17/1714* (2013.01); *A61B 17/1728* (2013.01); *A61B 17/1739* (2013.01); *A61B 17/1782* (2016.11); *A61B 2017/564* (2013.01); *A61F 2/0811* (2013.01); *Y10S 411/92* (2013.01)

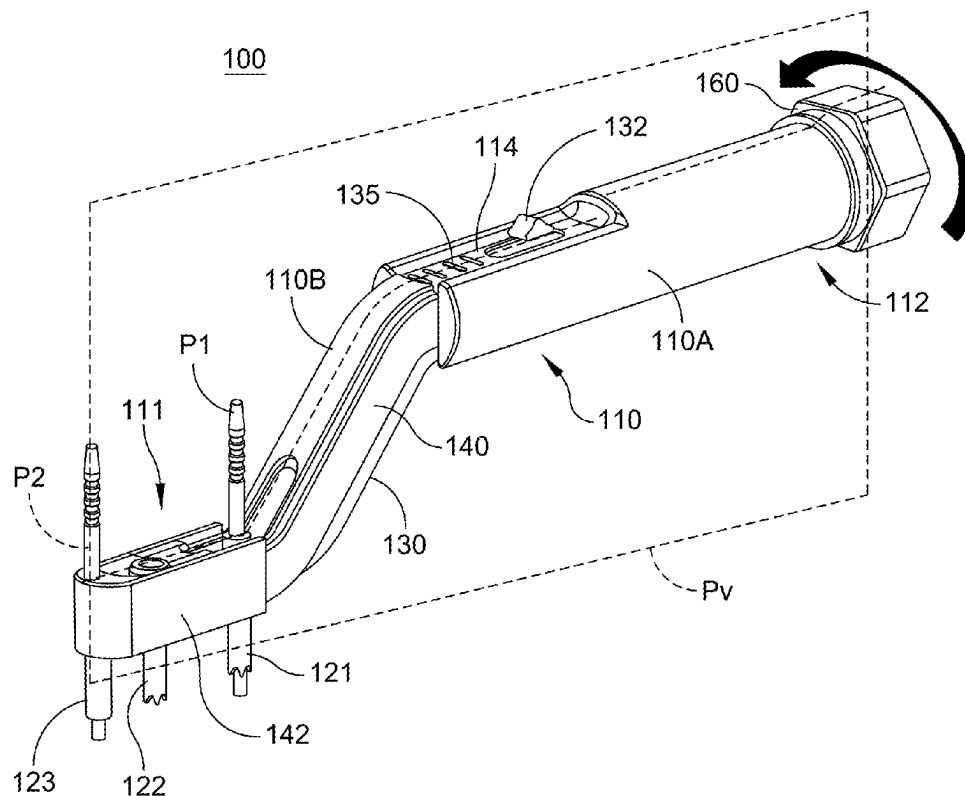
(21) Appl. No.: **19/181,411**(22) Filed: **Apr. 17, 2025****Related U.S. Application Data**

- (62) Division of application No. 17/759,567, filed on Jul. 27, 2022, filed as application No. PCT/US2021/018330 on Feb. 17, 2021, now Pat. No. 12,303,147.
- (60) Provisional application No. 63/001,910, filed on Mar. 30, 2020.

(57)

ABSTRACT

Novel orthopedic staples and related instruments are disclosed. An instrument is an orthopedic drill guide assembly that includes a first arm and a second arm that are retractably connected to the distal end of the instrument's outer housing, where the proximal ends of the arms are retractably connected to the distal end of the outer housing and configured to move in a retractable motion that changes the spacing between the first drill guide sleeve and the second drill guide sleeve, where the proximal end of the outer housing is configured for controlling the retractable motion of the first and second arms.



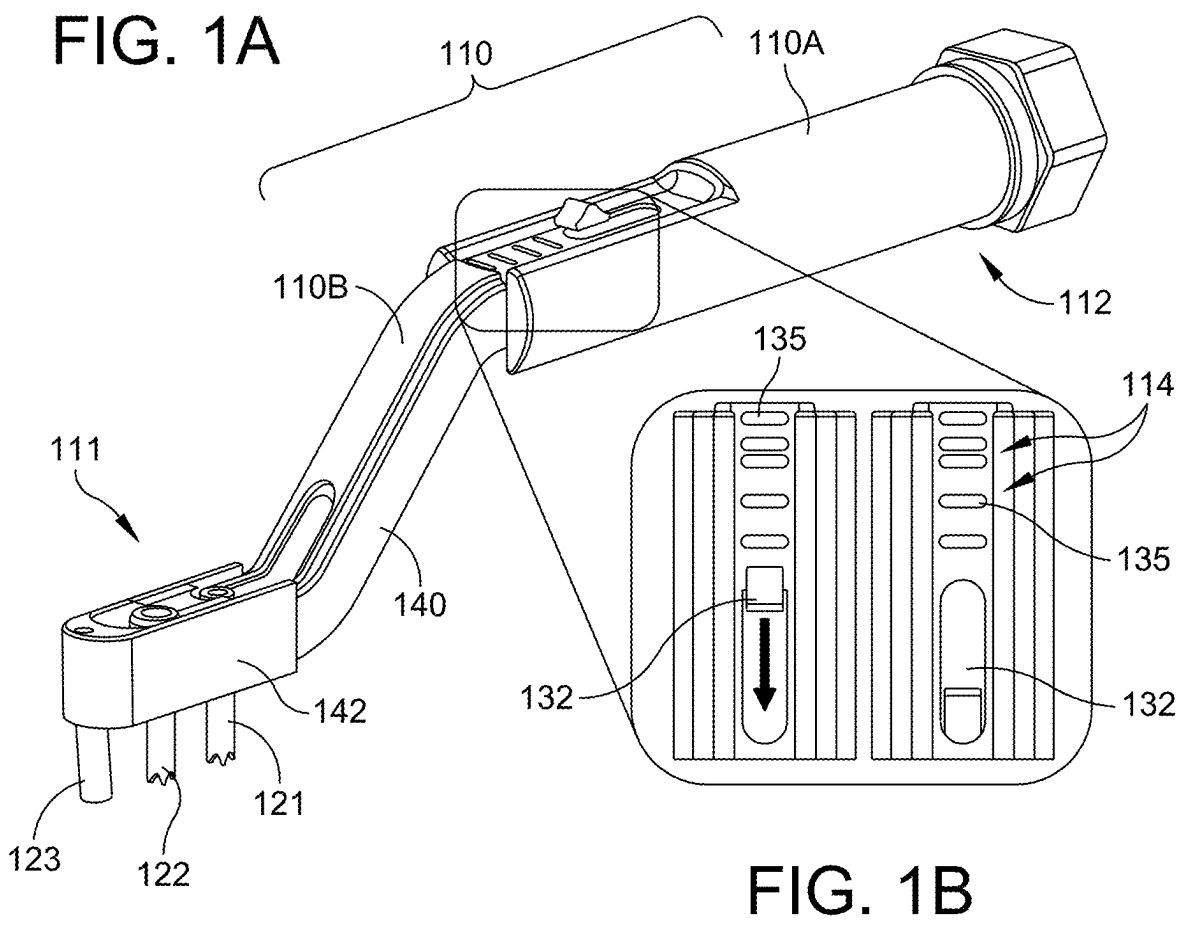


FIG. 2A

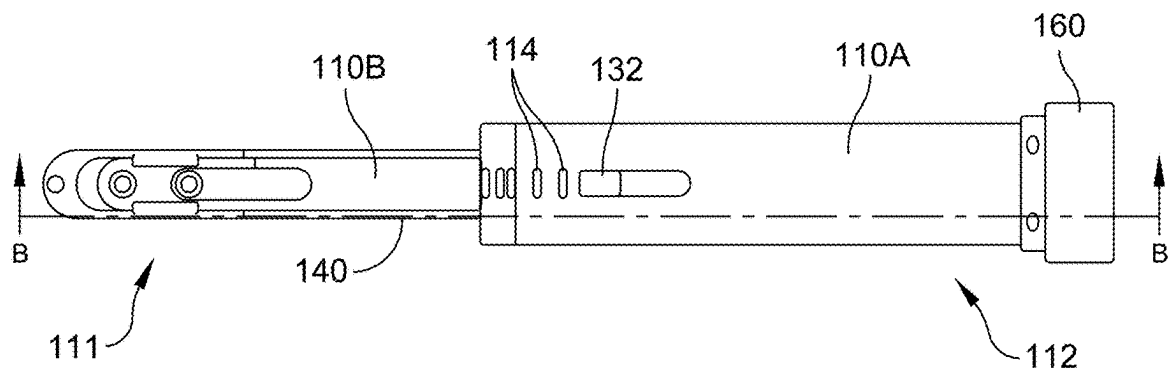


FIG. 2B

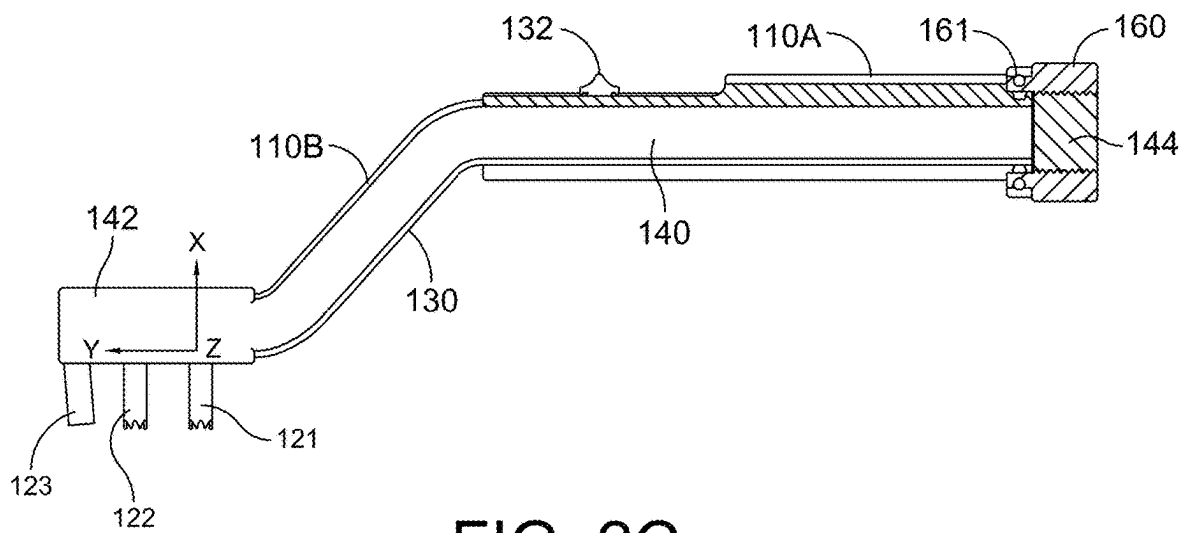
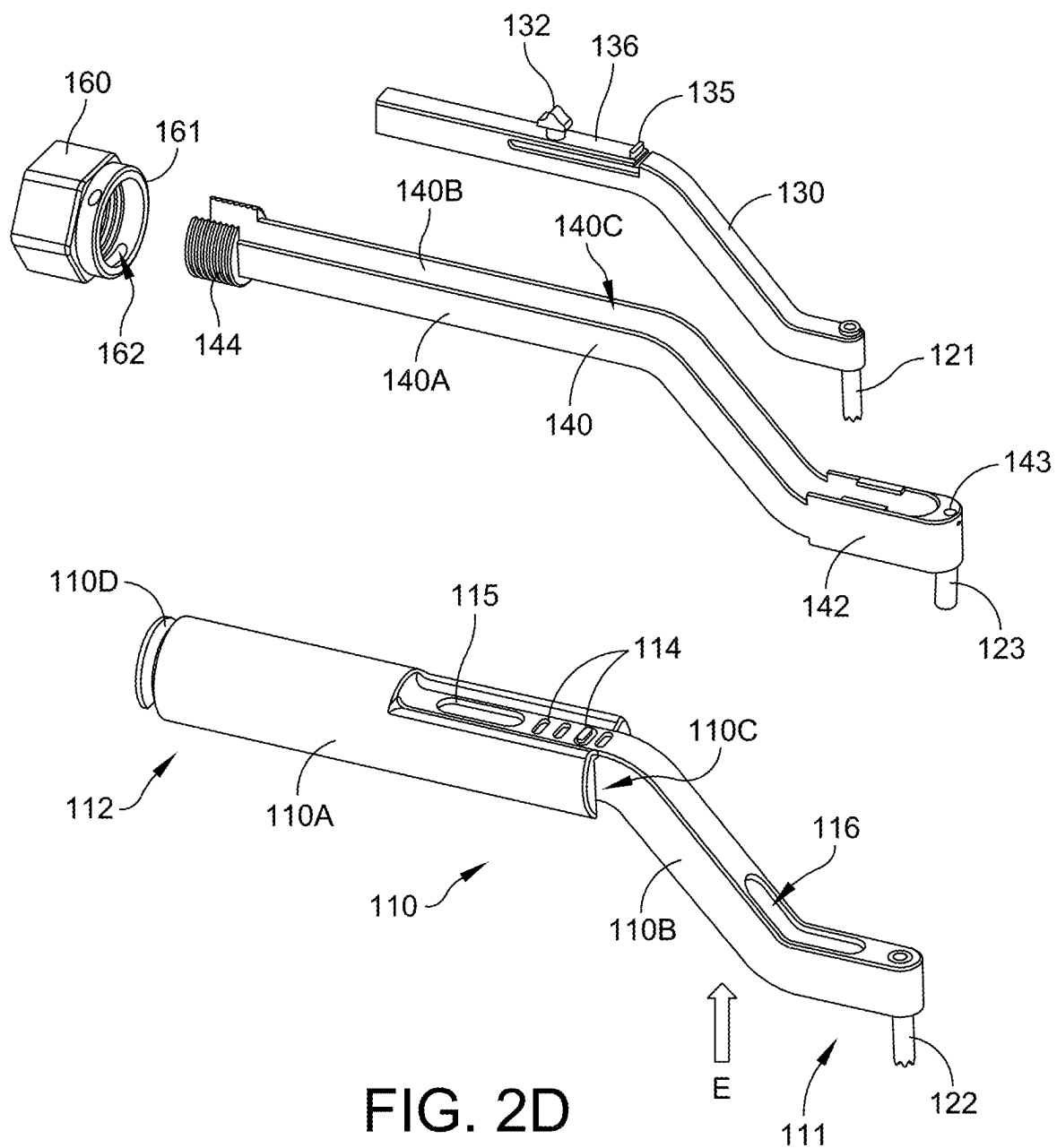


FIG. 2C



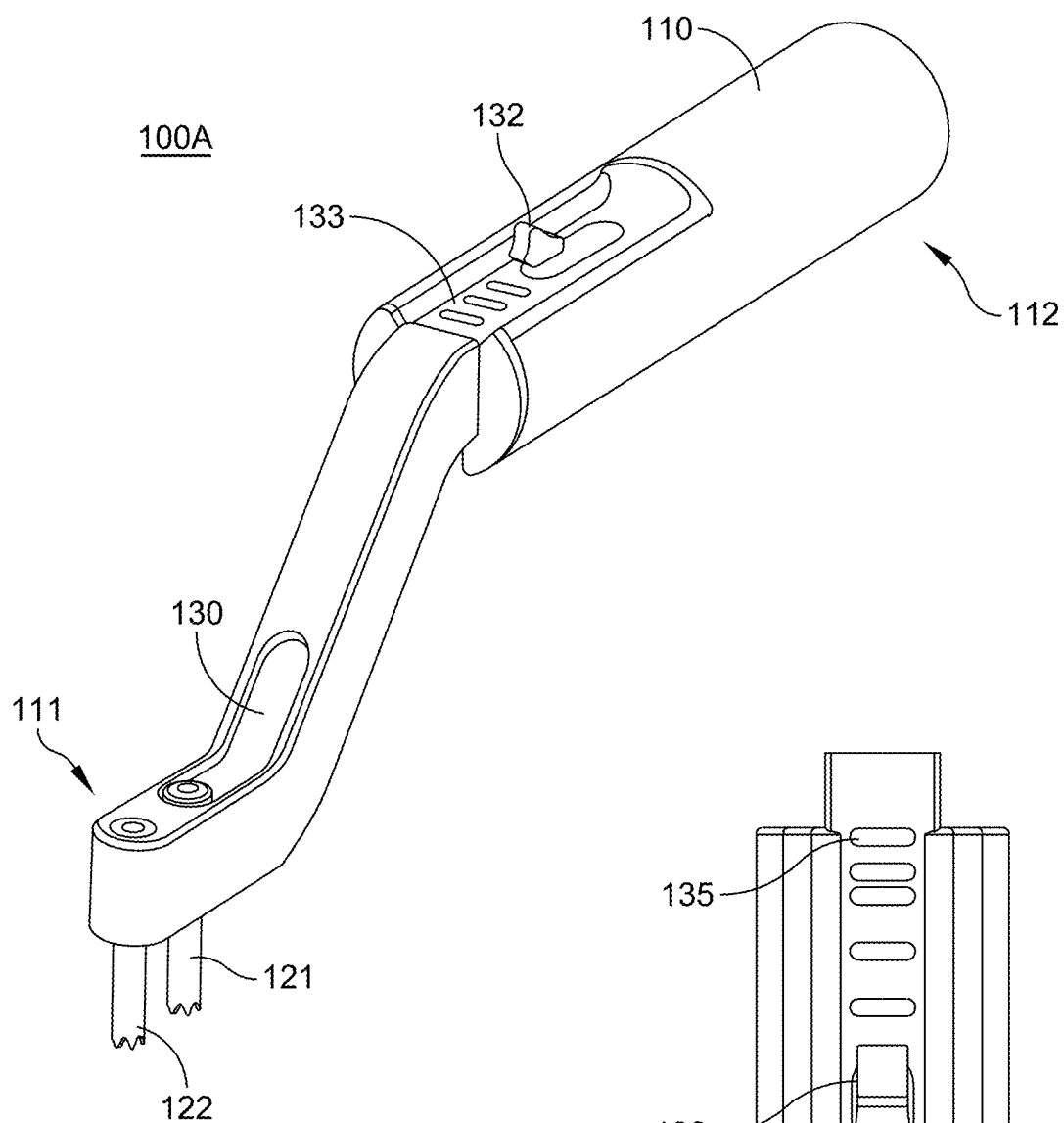


FIG. 3A

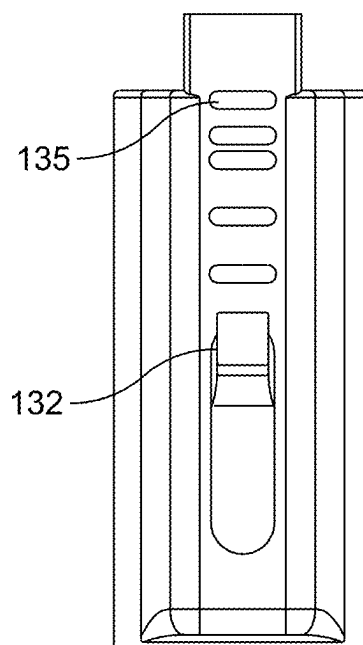


FIG. 3B

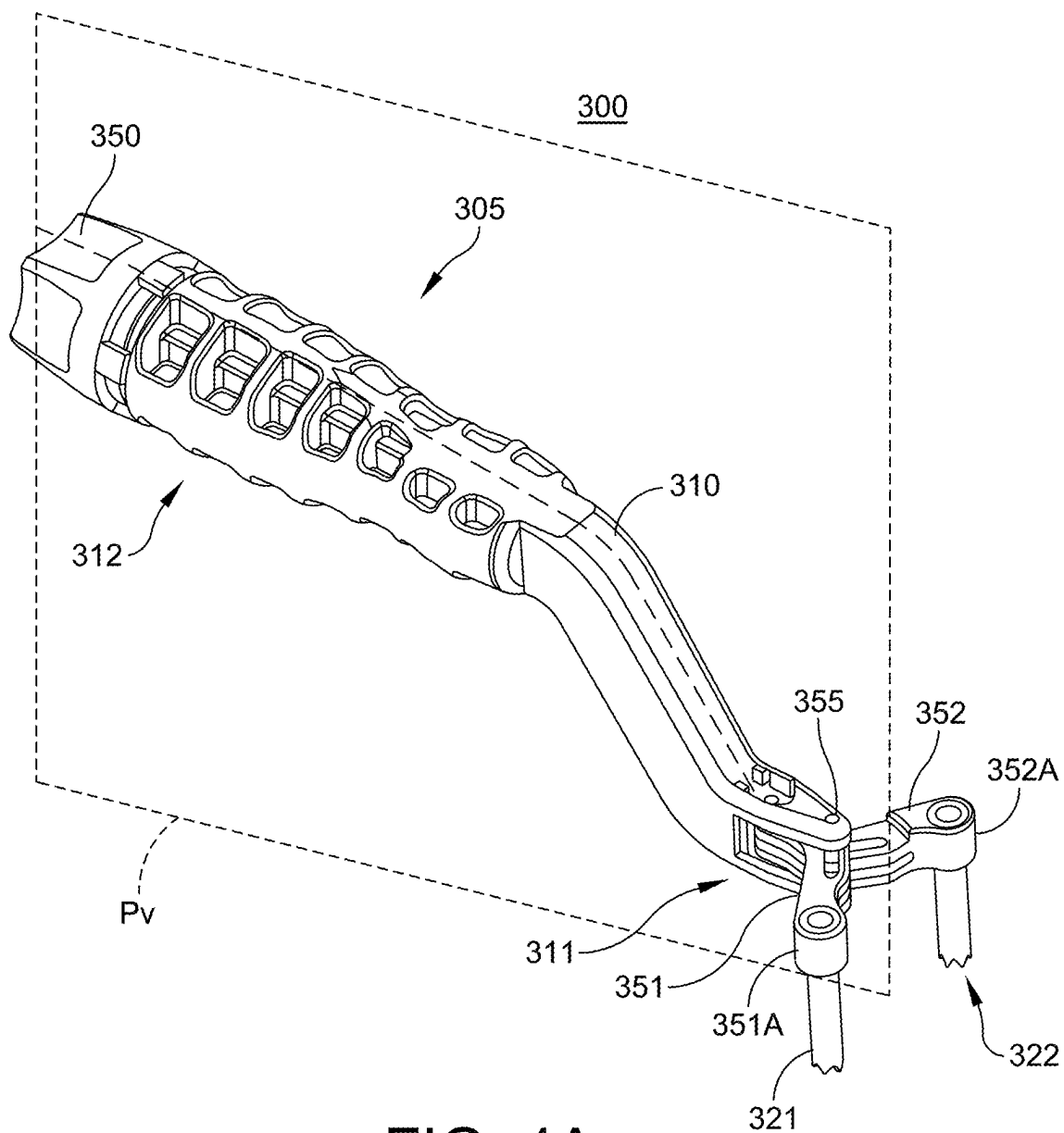


FIG. 4A

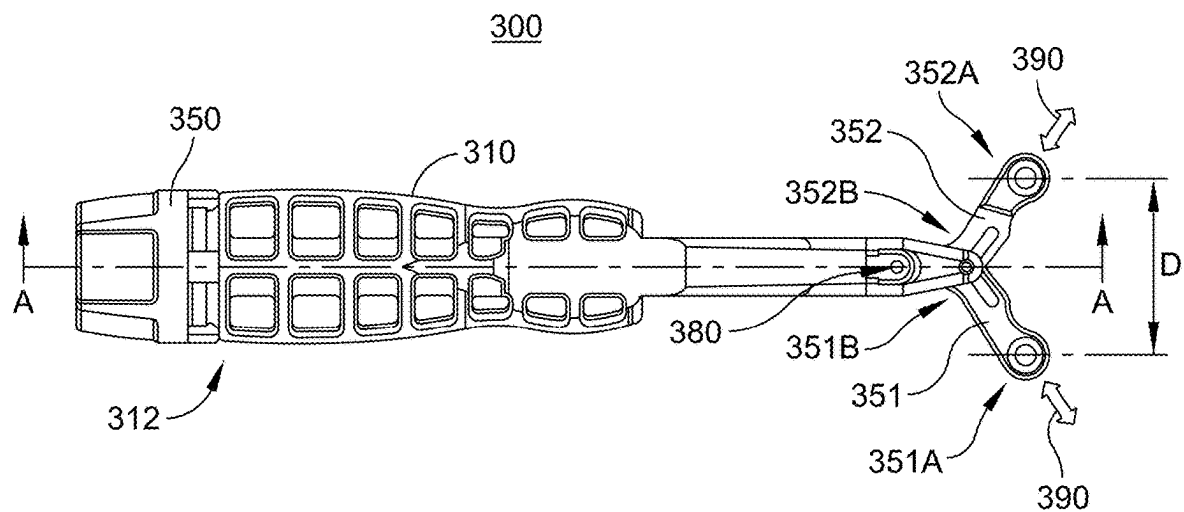


FIG. 4B

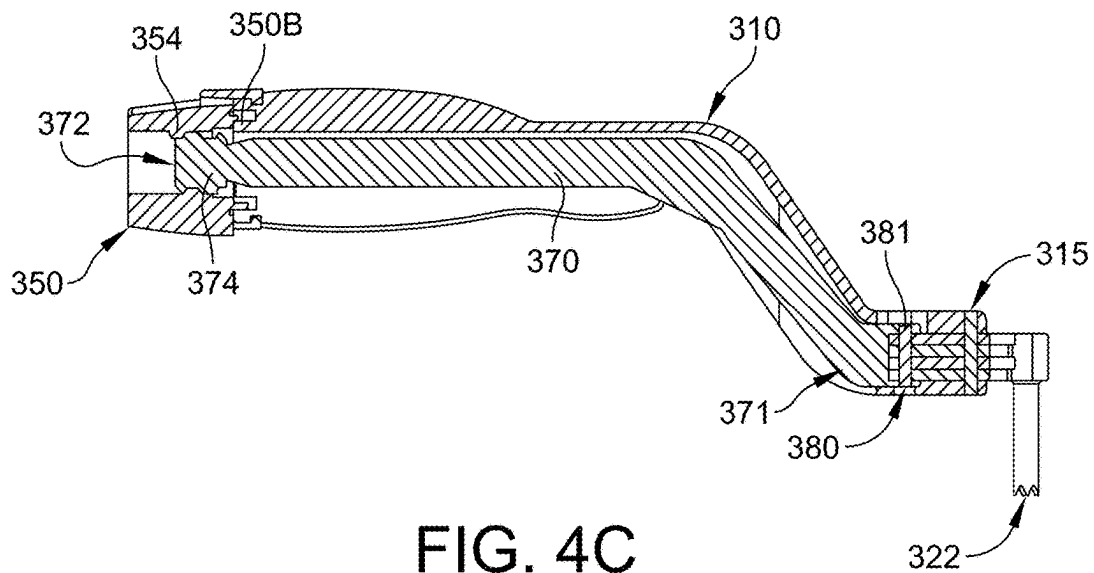


FIG. 4C

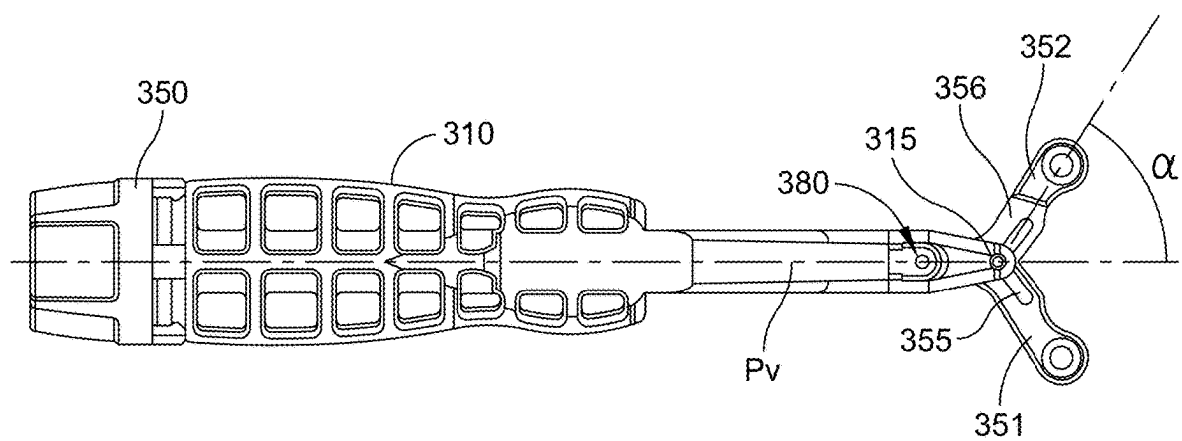


FIG. 4D

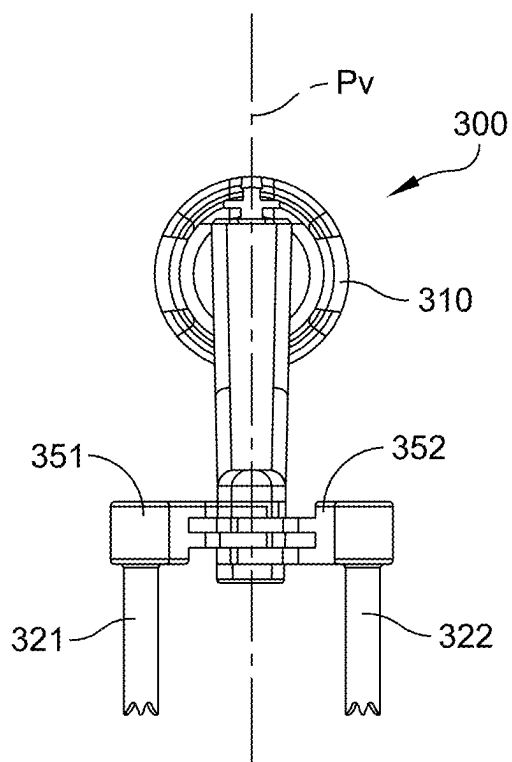


FIG. 4E

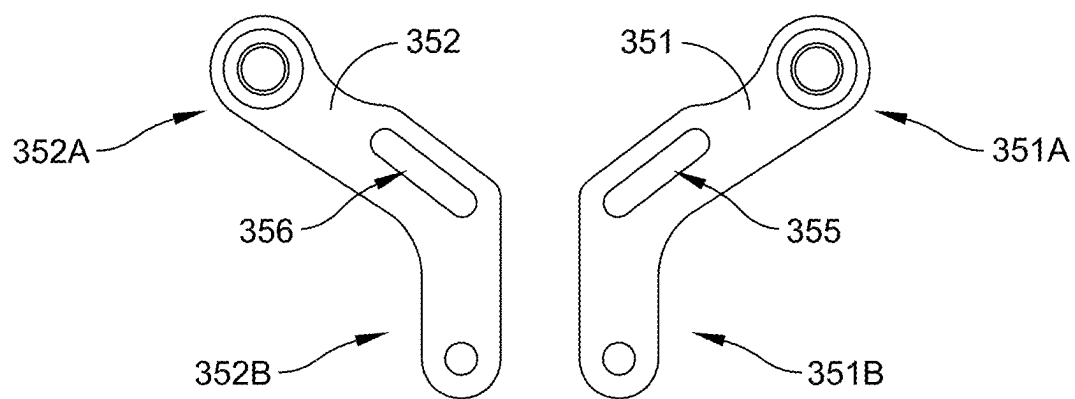
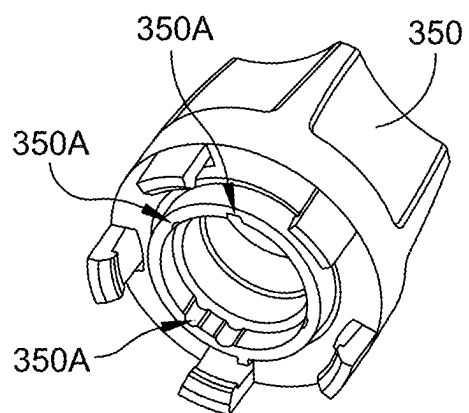
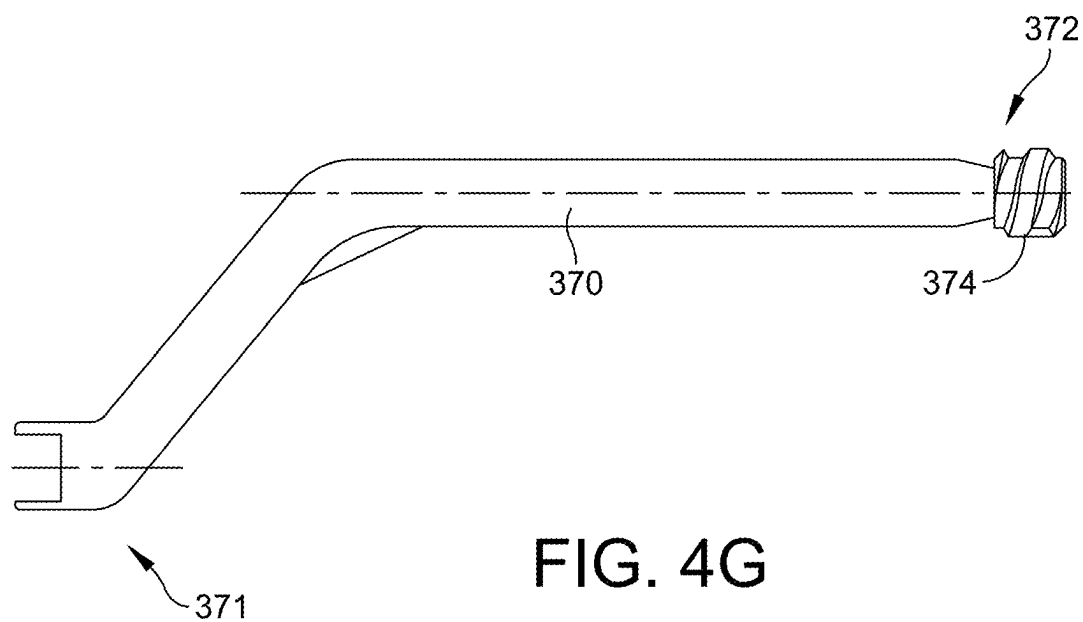


FIG. 4F



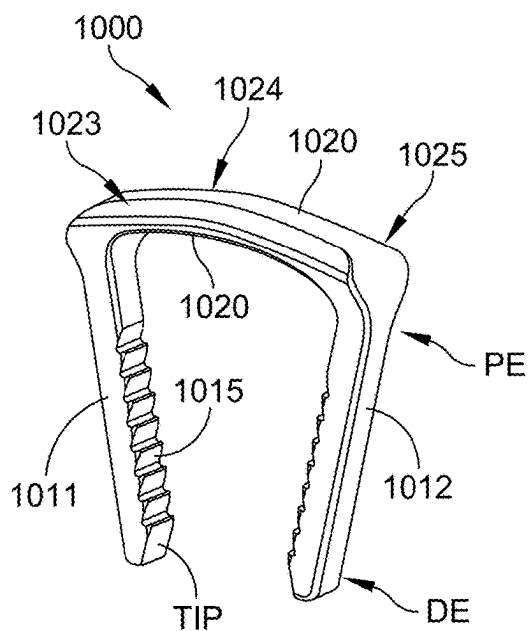


FIG. 5A

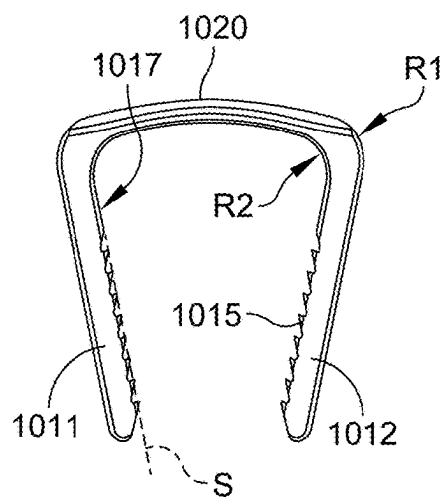


FIG. 5B

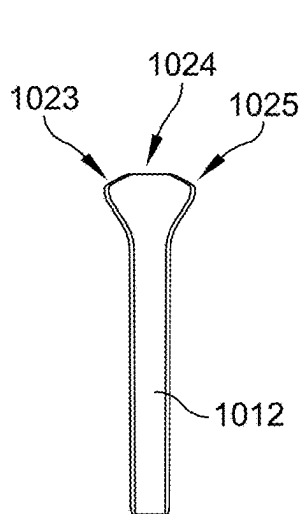


FIG. 5C

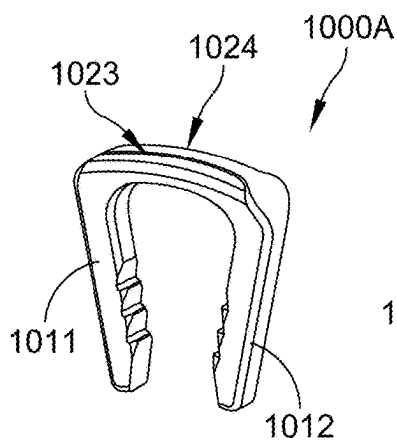


FIG. 5D

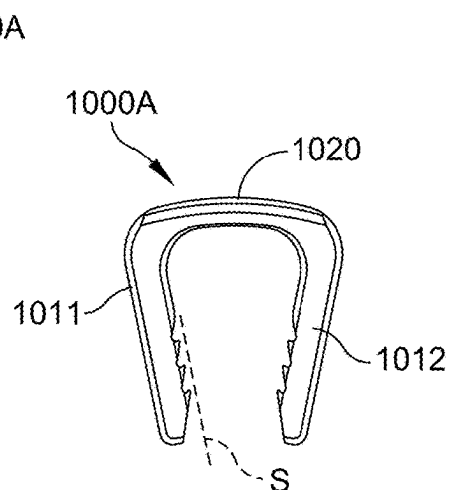


FIG. 5E

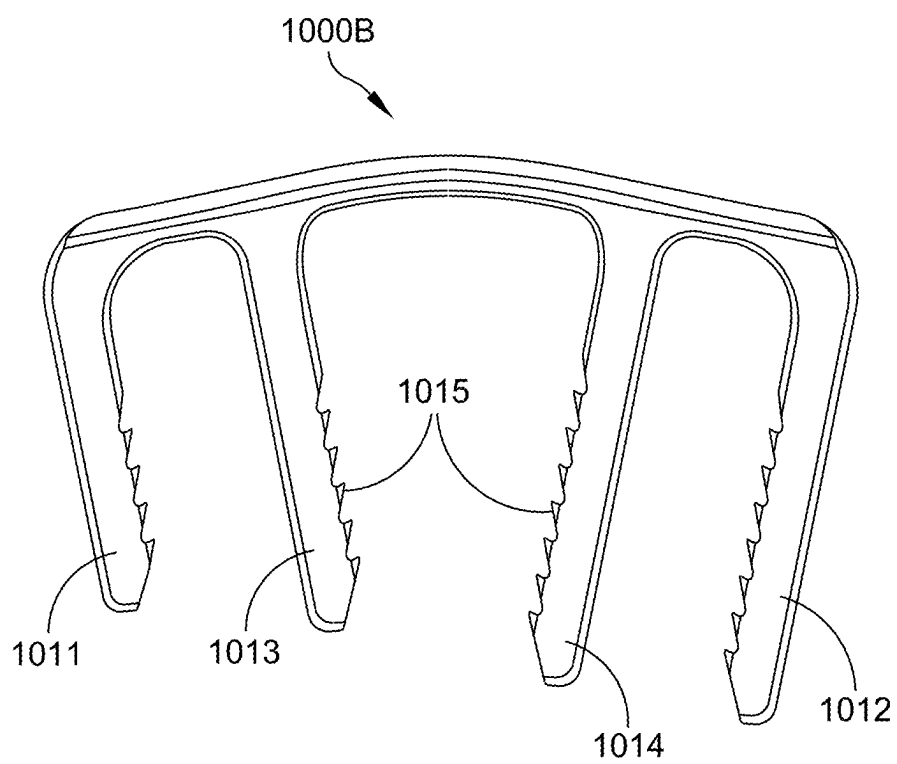


FIG. 5F

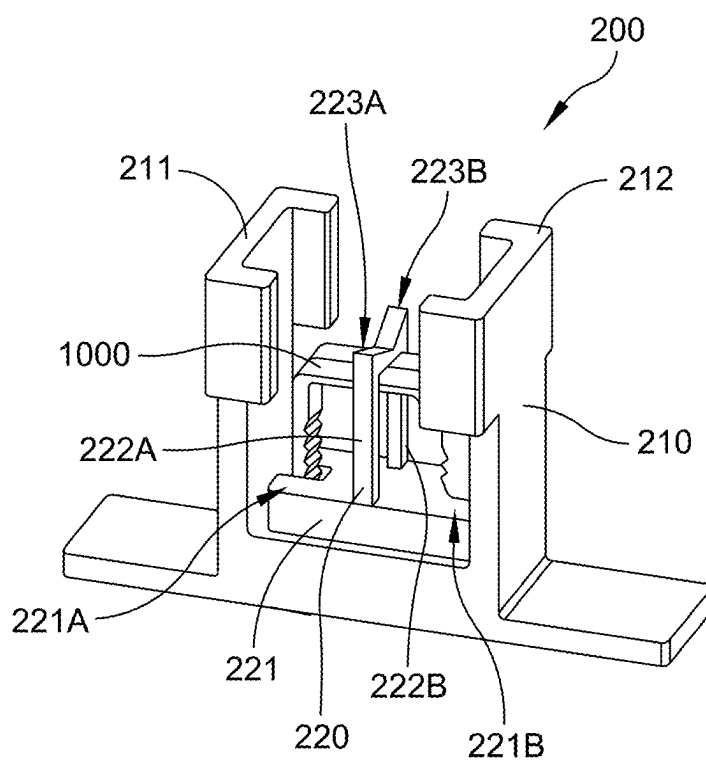


FIG. 6A

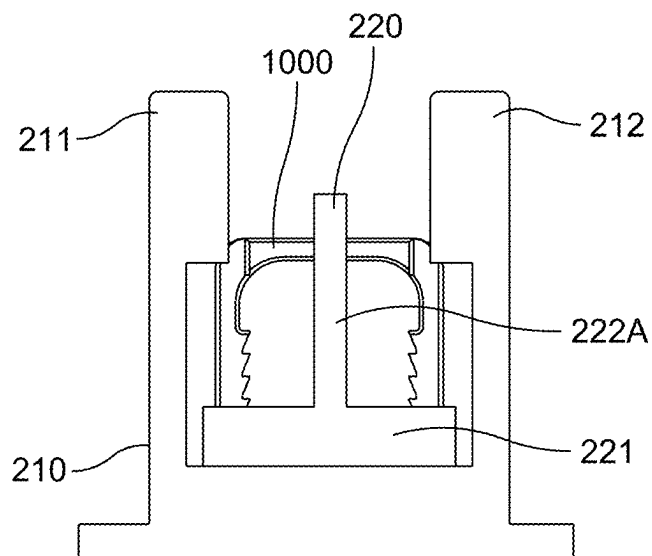


FIG. 6B

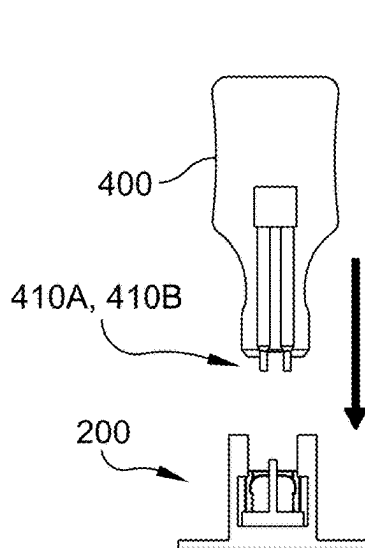


FIG. 7A

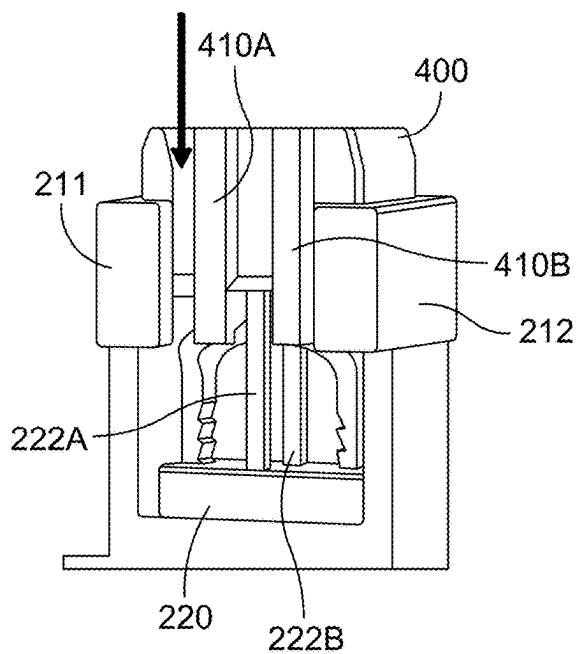


FIG. 7B

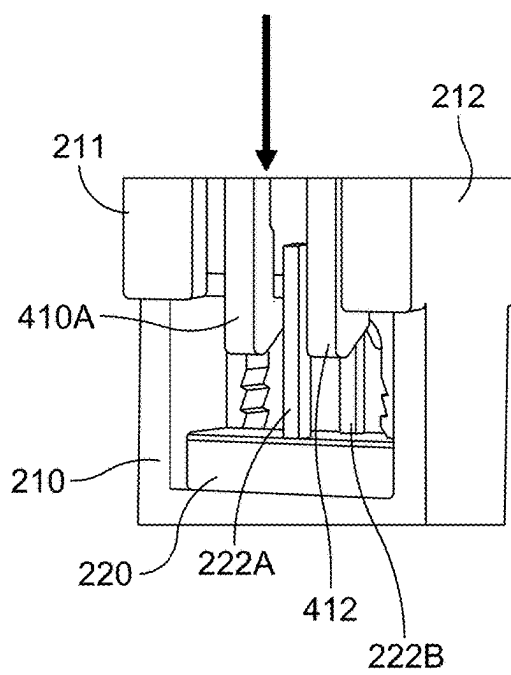


FIG. 7C

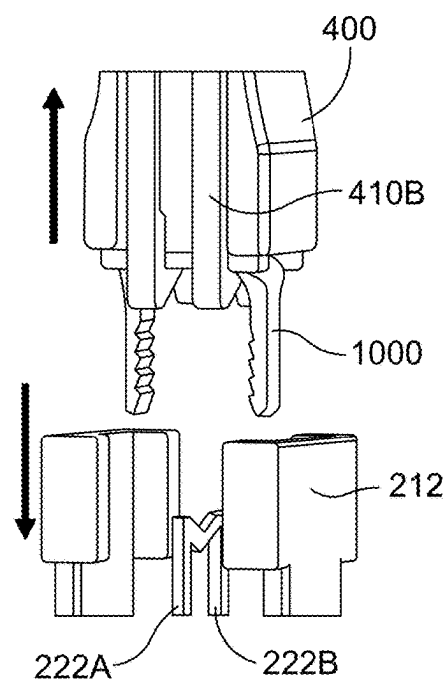


FIG. 7D

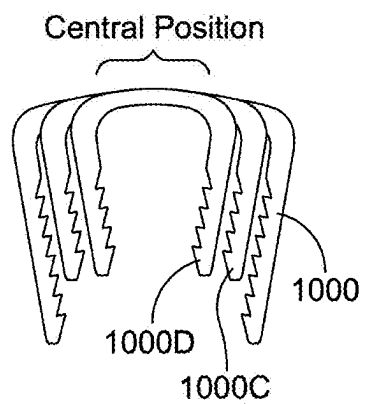


FIG. 8

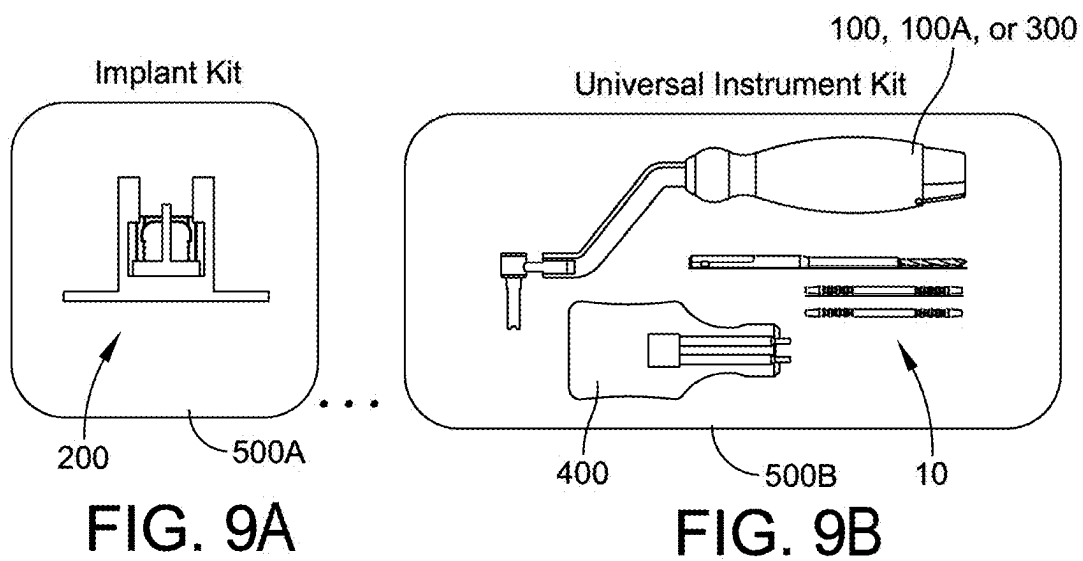


FIG. 9A

FIG. 9B

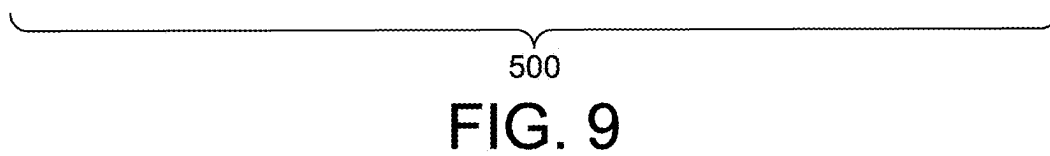


FIG. 9

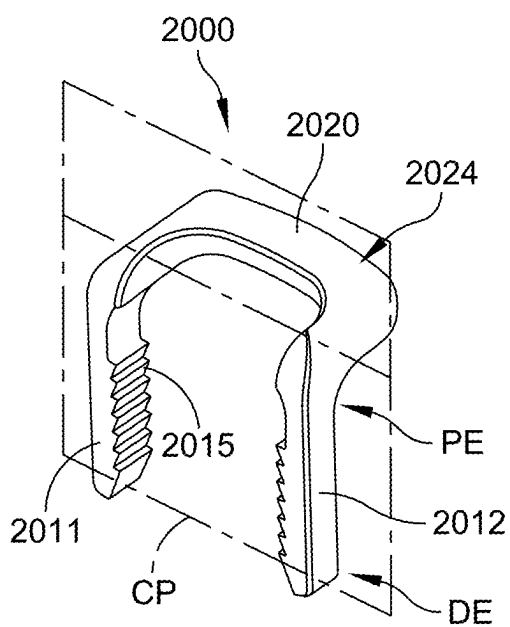


FIG. 10A

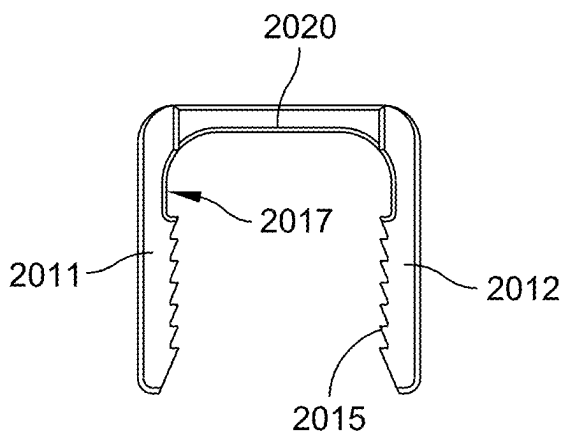


FIG. 10B

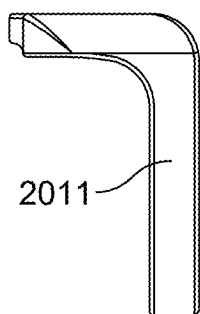


FIG. 10C

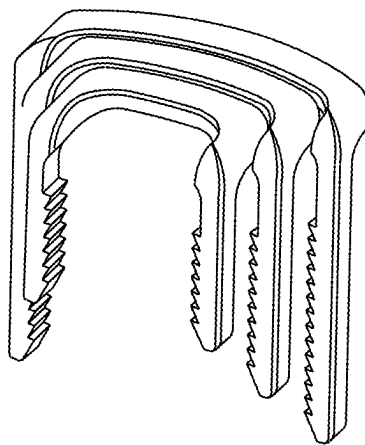


FIG. 10D

FIG. 11A

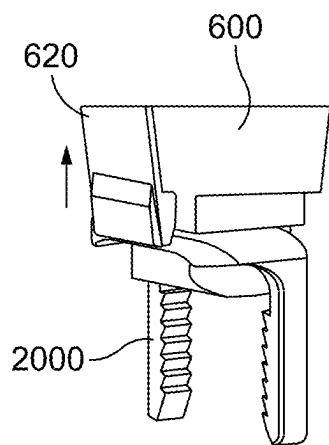
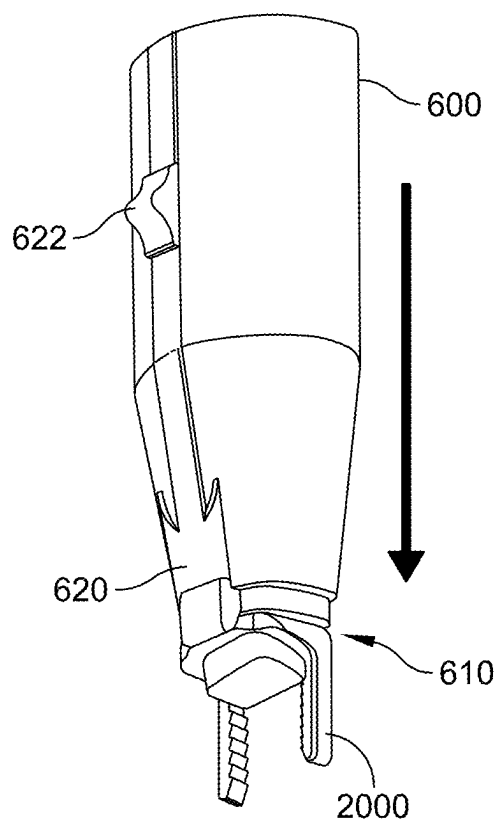


FIG. 11B

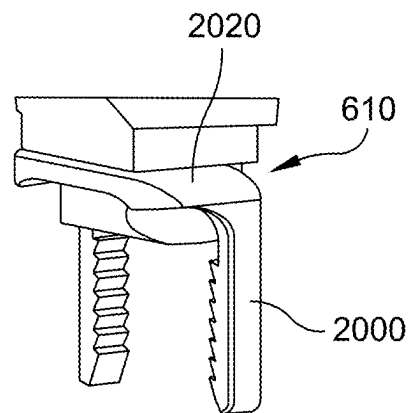


FIG. 11C

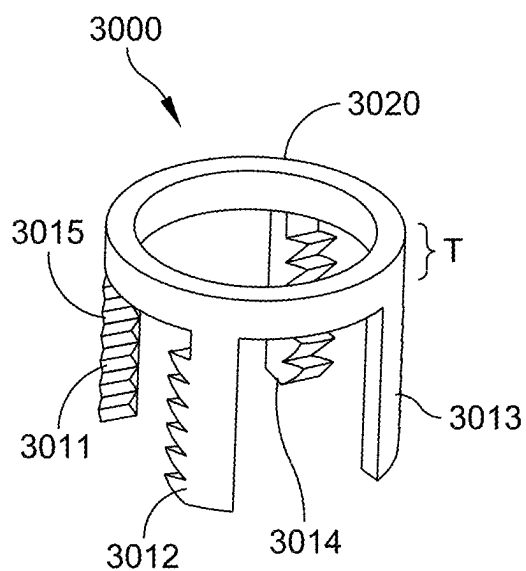


FIG. 12A

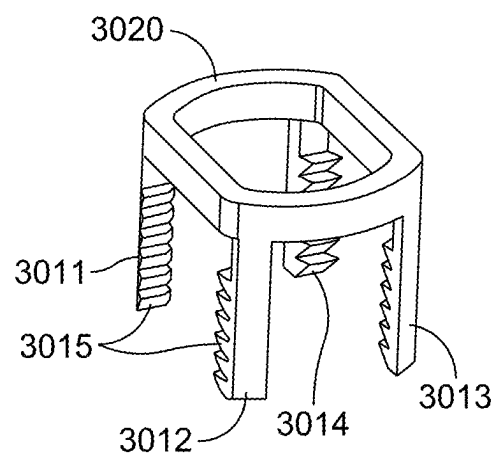


FIG. 12B

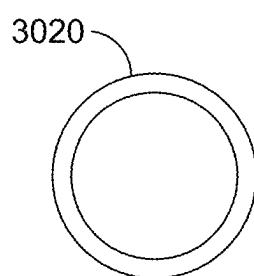


FIG. 12C

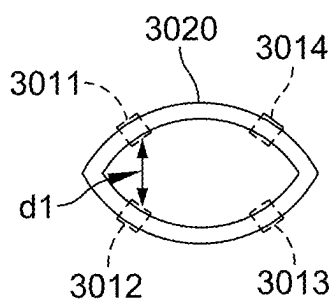


FIG. 12D

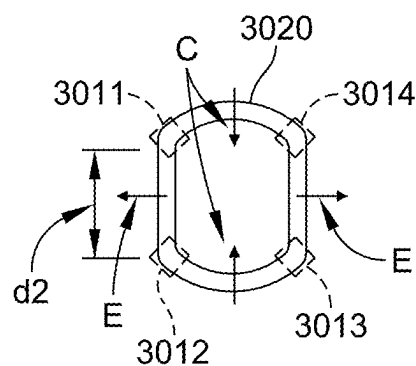


FIG. 12E

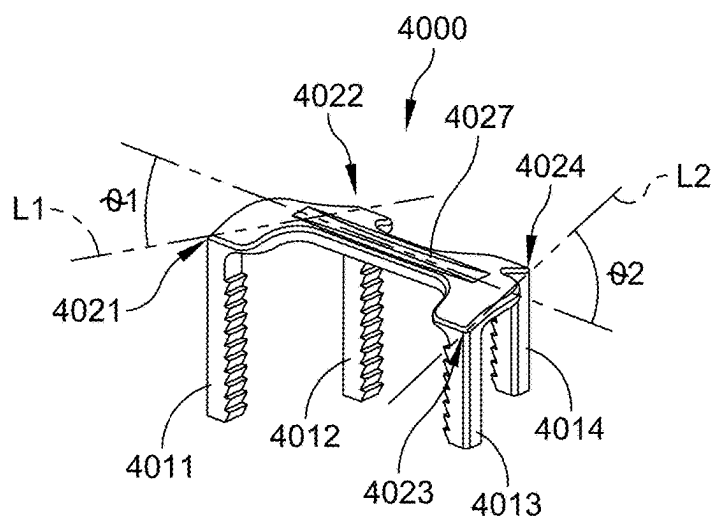


FIG. 13A

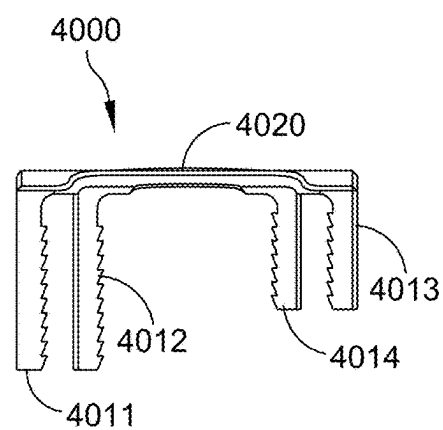


FIG. 13B

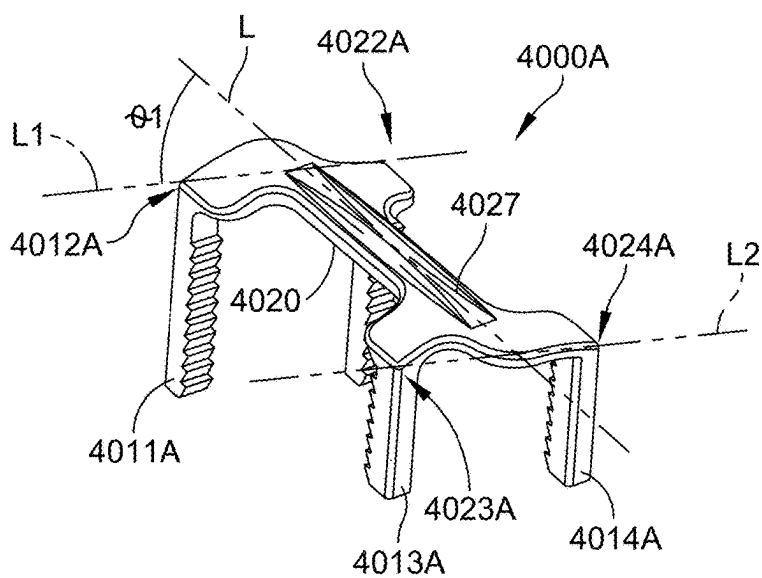


FIG. 13D

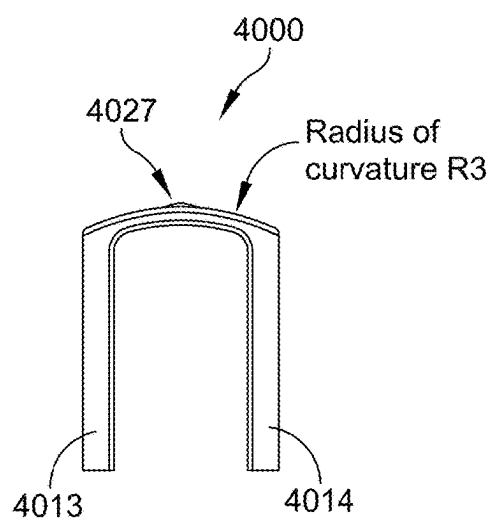


FIG. 13C

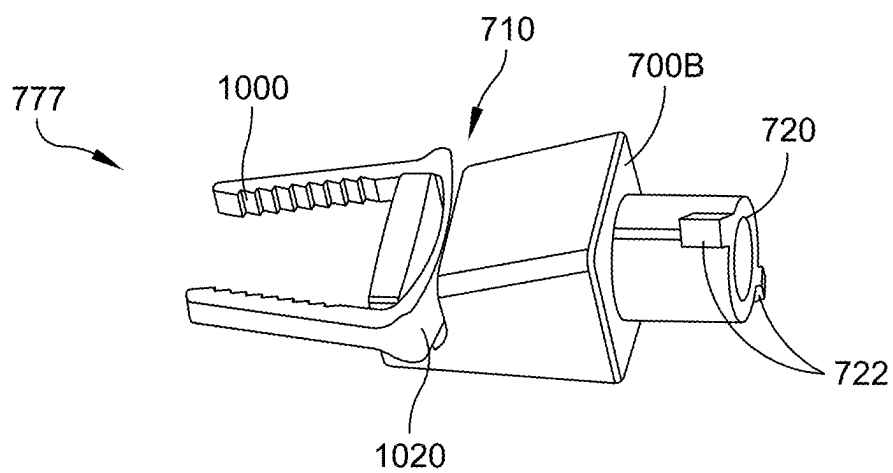
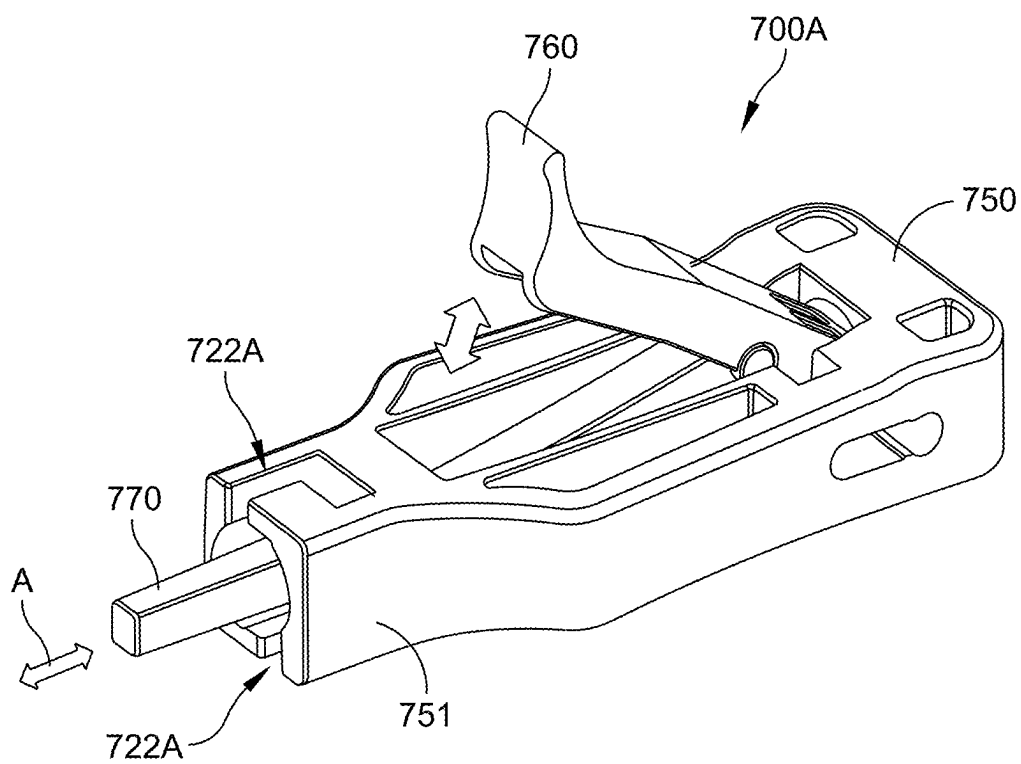


FIG. 14A



(Cartridge Receiving Position)

FIG. 14B

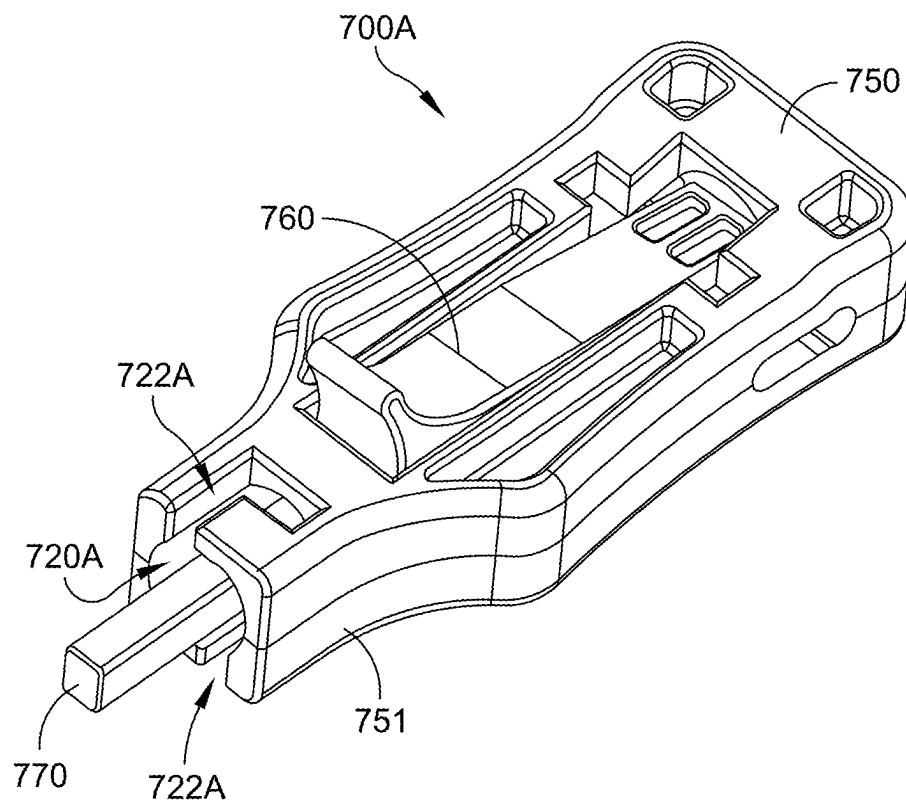


FIG. 14C

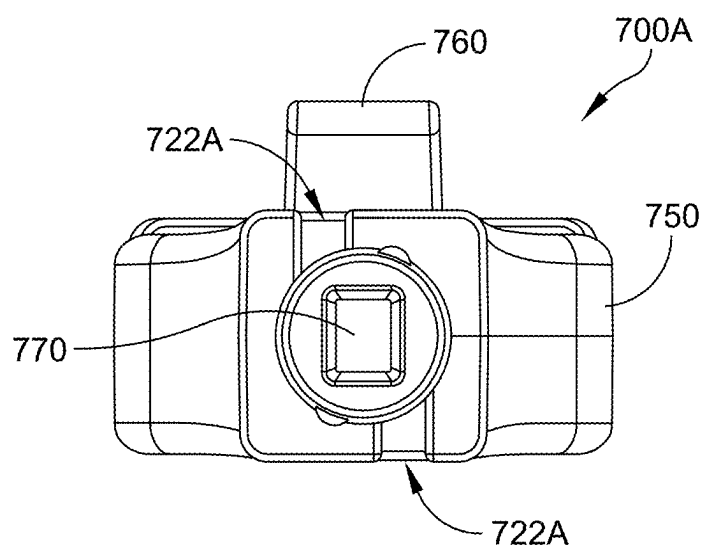
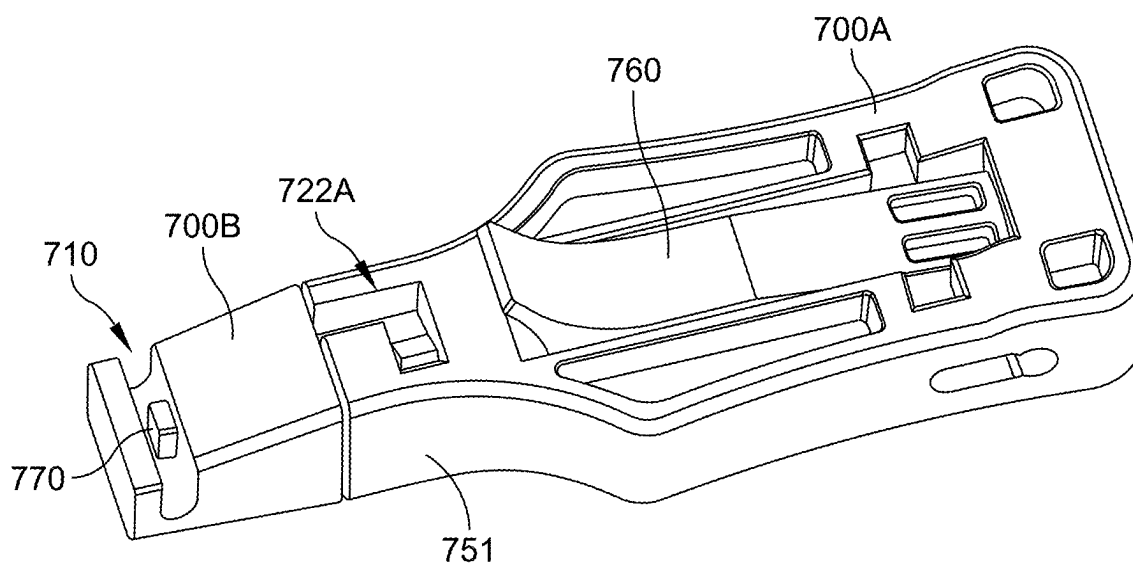


FIG. 14D



(Staple Inserting Position)

FIG. 14E

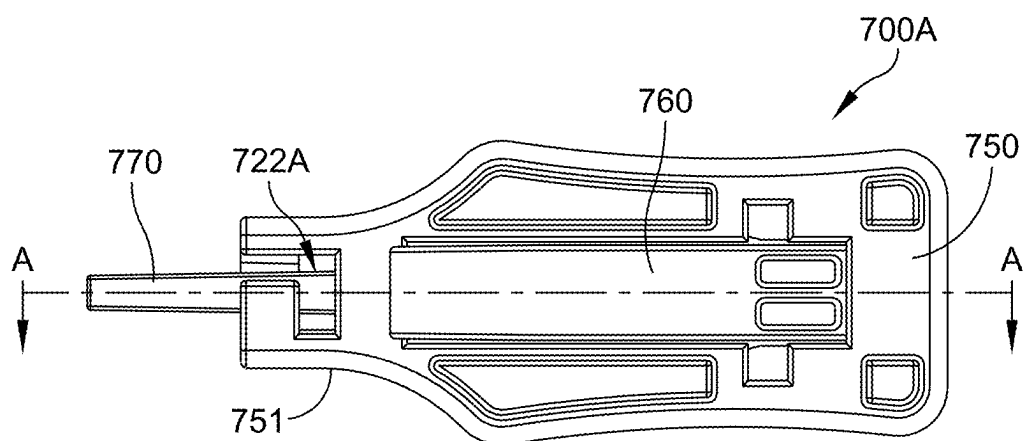
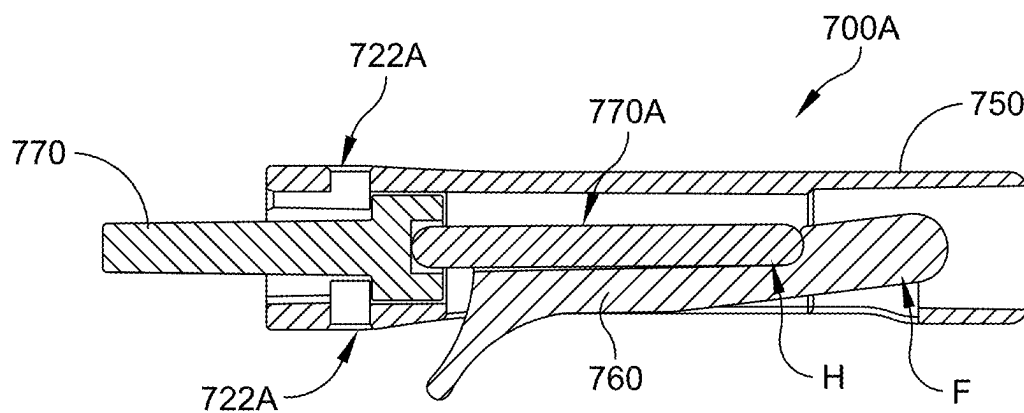


FIG. 14F



SECTION A-A

FIG. 14G

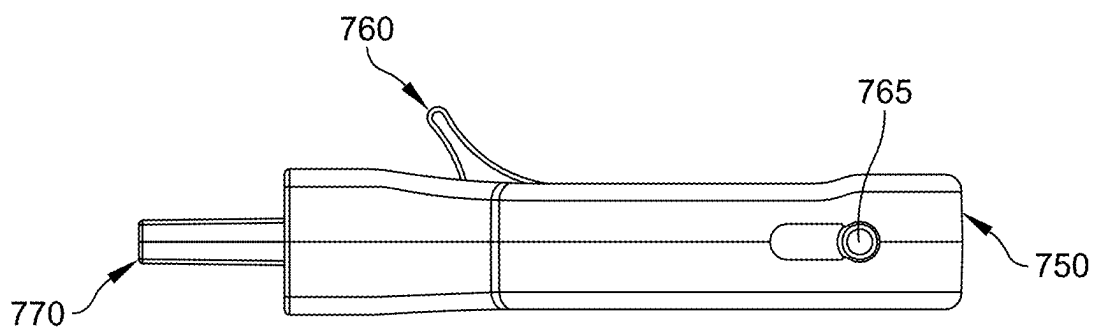


FIG. 14H

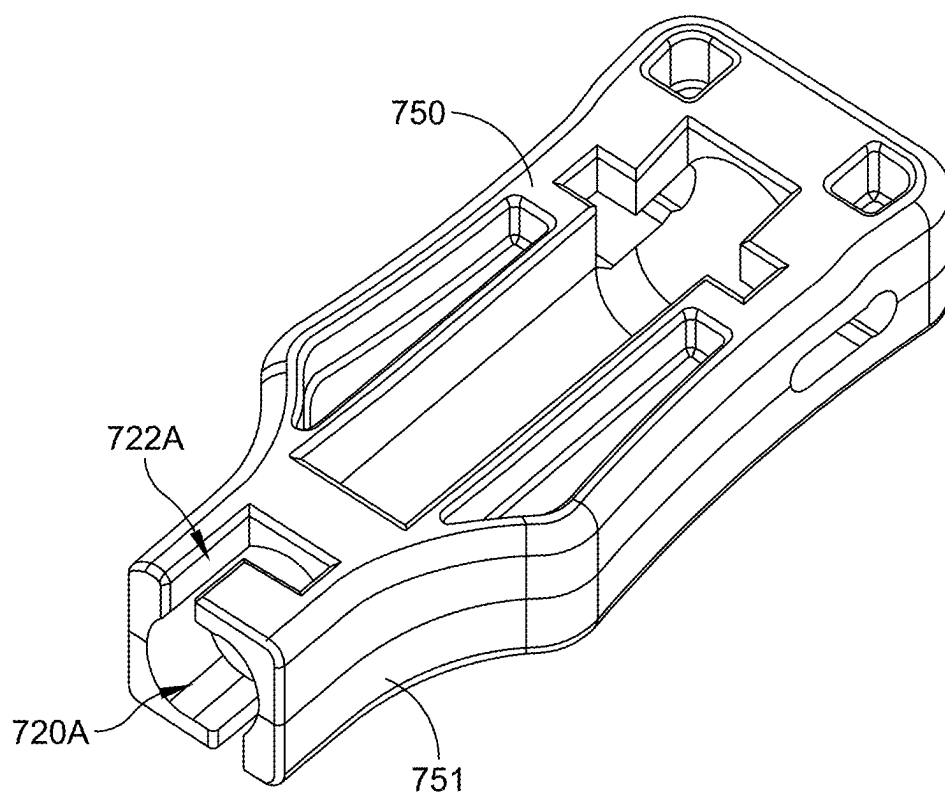


FIG. 14I

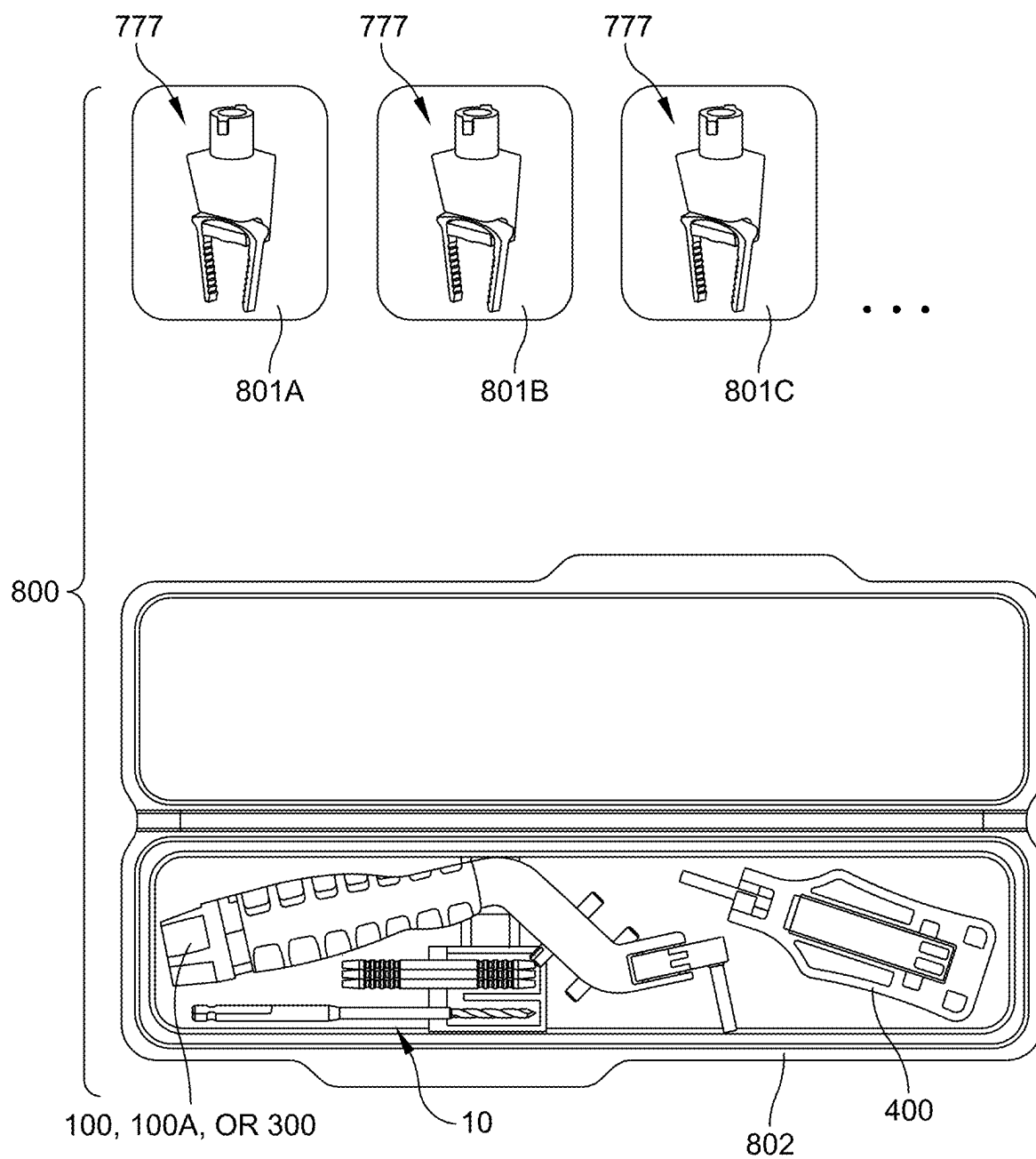


FIG. 15

ORTHOPEDIC STAPLE AND RELATED INSTRUMENTS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference in accordance with 37 C.F.R. §§ 1.57; 1.97; and 1.98.

FIELD OF DISCLOSURE

[0002] The present disclosure relates generally to orthopedic staples.

BACKGROUND

[0003] Orthopedic staples are utilized for moving and/or maintaining bones or bone parts substantially adjacent to one another after certain surgical procedures. Moreover, orthopedic staples are often utilized to compress and hold together in place two displaced pieces of bone while healing. Often a hole or holes are necessary in the bone material for proper insertion of the surgical staples.

SUMMARY

[0004] Disclosed herein are novel orthopedic staples and related instruments.

[0005] Provided is an orthopedic drill guide assembly comprising: an elongated body having a distal end, a proximal end, and a longitudinal vertical plane defined therethrough, wherein the elongated body comprising: an outer housing having a proximal end and a distal end; and a connecting piece provided within the outer housing; a first arm and a second arm retractably connected to the distal end of the outer housing, wherein each of the first arm and the second arm comprises a distal end and a proximal end; a first drill guide sleeve located at the distal end of the first arm; and a second drill guide sleeve located at the distal end of the second arm; wherein the first drill guide sleeve and the second drill guide sleeve are separated by a spacing, wherein the proximal ends of the arms are retractably connected to the distal end of the outer housing and configured to move in a retractable motion that changes the spacing between the first drill guide sleeve and the second drill guide sleeve, wherein the proximal end of the outer housing is configured for controlling the retractable motion of the first and second arms.

[0006] Also provided is an orthopedic drill guide assembly comprising: an elongated body having a distal end and a proximal end, and a longitudinal vertical plane defined therethrough; a first guide hole and a second guide hole located near the distal end of the elongated body, wherein the first and second guide holes are arranged in a linear alignment with the longitudinal vertical plane of the elongated body such that the first guide hole is between the second guide hole and the proximal end of the elongated body; a distal sliding portion slidably engaging the distal end of the elongated body and is configured to slide back and forth along a direction that is in-line with the longitudinal vertical plane of the elongated body, wherein the distal sliding portion is communicatively connected to the proximal end of the elongated body and the distal sliding portion's sliding motion is actuated and controlled from the

proximal end of the elongated body; and a third guide hole located on the distal sliding portion and in-line with the first and second guide holes along the longitudinal vertical plane of the elongated body, wherein the first and second guide holes are located between the third guide hole and the proximal end, whereby the distal sliding portion's sliding motion changes the distance between the third guide hole and the first guide hole.

[0007] Also provided is an orthopedic drill guide assembly comprising: an elongated body having a distal end and a proximal end, and a longitudinal vertical plane defined therethrough; a first drill guide hole and a second drill guide hole located near the distal end of the elongated body, wherein the first and second drill guide holes are arranged in a linear alignment with the longitudinal vertical plane of the elongated body such that the first drill guide hole is between the second drill guide hole and the proximal end of the elongated body

[0008] Also provided is a bone staple comprising: a first leg; a second leg; and a bridge connecting the two legs, the bridge comprising a top surface along the length of the bridge, wherein the top surface comprises: two chamfered edge surfaces that extend substantially the length of the bridge; and one flat surface between the two chamfered edge surfaces, wherein the flat surface extends substantially the length of the bridge, wherein each of the first end and the second end of the bridge transitions down to a leg.

[0009] Also provided is a bone staple comprising: a pair of legs, each having a proximal end and a distal end, the pair of legs defining a compression plane; and a non-linearly extending bridge portion connecting the distal ends of the pair of legs, wherein the bridge portion has a generally U-shape and comprises: a flat portion that spans the distance between the distal ends of the pair of legs; and two ends of the U-shape connecting the flat portion to the proximal ends of the legs, wherein the two ends of the U-shape extending laterally between the flat portion and the proximal ends of the legs, whereby the flat portion is off-set from the compression plane; wherein the flat portion has a width that is greater than its thickness.

[0010] Also provided is a surgical staple comprising: a bridge that has an open loop configuration and a thickness from 0.05 upto 0.1 inches; two or more pairs of legs extending from the bridge in the same direction, wherein each of the legs has a surface that faces the other leg in its pair, wherein the surface comprises a plurality of barbs.

[0011] Also provided is a surgical staple comprising: a bridge having an elongated shape with a longitudinal axis defined through its length, the bridge further comprising first, second, third, and fourth corners; a first leg extending from the first corner; a second leg extending from the second corner; a third leg extending from the third corner; a fourth leg extending from the fourth corner; the bridge comprising a first section that includes the first and second corners and a second section that includes the third and fourth corners; and wherein the first and second corners are arranged such that a line connecting the first and second corners intersect the longitudinal axis of the bridge at a first non-orthogonal angle.

[0012] Also provided is a bone staple inserter kit comprising: a first sterile package containing a first bone staple securely mounted to a first cartridge, wherein the first bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion

connecting the first and second legs, wherein the first cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge portion of the first bone staple and securely hold the first bone staple by an interference fit; and a second sterile package containing; a bone staple insertion tool that is configured to receive a cartridge, like the first cartridge that has a bone staple, like the first bone staple, securely mounted thereon; a universal drill guide assembly; and a drill bit, wherein the bone staple insertion tool comprises an actuation mechanism that can transform the bone staple to an insertion shape after the cartridge with the bone staple mounted thereon is mounted on the insertion tool.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The various embodiments of the inventive hydrogel implant of the present disclosure will be described in more detail in conjunction with the following drawing figures. The structures in the drawing figures are illustrated schematically and are not intended to show actual dimensions.

[0014] FIGS. 1A-2D illustrate an embodiment of an orthopedic drill guide assembly for predrilling holes in two bone pieces for orthopedic staples.

[0015] FIGS. 3A and 3B illustrate another embodiment of an orthopedic drill guide assembly for predrilling holes in two bone pieces for orthopedic staples.

[0016] FIGS. 4A to 4H illustrate another embodiment of an orthopedic drill guide assembly for predrilling holes in two bone pieces for orthopedic staples.

[0017] FIGS. 5A-5F are illustrations of shape memory metal staples having a chamfered bridge according to an embodiment.

[0018] FIGS. 6A and 6B illustrate a universal staple holder for inline bone staples according to an embodiment of the present disclosure.

[0019] FIGS. 7A to 7D illustrate the universal bone staple holder of FIGS. 6A, 6B interacting with a universal staple inserter for inline bone staples according to an embodiment of the present disclosure.

[0020] FIG. 8 is an illustration of an overlapping view of three different size staples that are configured for using with the universal staple holder shown in FIGS. 6A and 6B.

[0021] FIGS. 9A and 9B in combination show a surgical kit providing inline bone staples in the universal staple holders shown in FIG. 6A and the associated universal instruments: the orthopedic drill guide shown in FIG. 4A; and a universal staple inserter shown in FIG. 7A.

[0022] FIGS. 10A-10D are illustrations of shape memory metal staples having an offset bridge according to an embodiment.

[0023] FIGS. 11A-11C are illustrations showing a staple inserter for the offset bridge staples of FIGS. 10A-10D.

[0024] FIGS. 12A-12E are illustrations showing another shape memory metal staple according to another embodiment.

[0025] FIGS. 13A-13D are illustrations showing another shape memory metal staple according to another embodiment.

[0026] FIGS. 14A-14I are illustrations showing a universal staple inserter system of the present disclosure.

[0027] FIG. 15 is an illustration of a surgical kit according to the present disclosure.

DETAILED DESCRIPTION

[0028] This description of the exemplary embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. The drawing figures are not necessarily to scale and certain features may be shown exaggerated in scale or in somewhat schematic form in the interest of clarity and conciseness. In the description, relative terms such as “horizontal,” “vertical,” “up,” “down,” “top” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing figure under discussion. These relative terms are for convenience of description and normally are not intended to require a particular orientation. Terms including “inwardly” versus “outwardly,” “longitudinal” versus “lateral” and the like are to be interpreted relative to one another or relative to an axis of elongation, or an axis or center of rotation, as appropriate. Terms concerning attachments, coupling and the like, such as “connected” and “interconnected,” refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. When only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. The term “operatively connected” is such an attachment, coupling or connection that allows the pertinent structures to operate as intended by virtue of that relationship. In the claims, means-plus-function clauses, if used, are intended to cover the structures described, suggested, or rendered obvious by the written description or drawings for performing the recited function, including not only structural equivalents but also equivalent structures.

[0029] Referring to FIGS. 1A-2, an embodiment of an orthopedic drill guide assembly **100** for predrilling holes in two bone pieces for orthopedic staples is disclosed. The orthopedic drill guide assembly **100** comprises an elongated body formed by a first housing **110** having a distal end **111** and a proximal end **112**, and a longitudinal vertical plane Pv defined therethrough. The first housing **110** comprises a handle portion **110A** and a neck portion **110B**. A first guide hole **121** and a second guide hole **122** are located near the distal end **111** of the elongated body. The first and second guide holes **121**, **122**, respectively, are arranged in a linear alignment with the longitudinal vertical plane Pv of the elongated body such that the first guide hole **121** is between the second guide hole **122** and the proximal end **112** of the elongated body.

[0030] The first and second guide holes **121**, **122** are the drill guides that are used to predrill holes into bone repair site where an orthopedic staple will be placed. Therefore, the distance between the first and second guide holes **121**, **122** is set to represent the base distance between the two legs of an orthopedic staple. The term “base distance” refers to the distance between the two legs of an orthopedic staple that is in its insertion configuration. In other words, the state where the two legs of the staple are parallel to each other and ready to be inserted into a bone repair site.

[0031] At the distal end **111** is provided a distal sliding portion **142** that slidably engages the distal end of the elongated body and is configured to slide back and forth

along a direction that is in-line with the longitudinal vertical plane of the elongated body. The distal sliding portion **142** is communicatively connected to the proximal end **112** of the elongated body and the distal sliding portion's sliding motion is actuated and controlled from the proximal end **112** of the drill guide assembly **100**. The elongated body also comprises a third guide hole **123** located on the distal sliding portion **142** and in-line with the first **121** and second **122** guide holes along the longitudinal vertical plane **Pv** of the elongated body. The first and second guide holes **121**, **122** are located between the third guide hole **123** and the proximal end **112**, whereby the distal sliding portion's sliding motion changes the distance between the third guide hole **123** and the first guide hole **121**. The distal sliding portion **142** is part of a second housing **140** that connects the distal sliding portion **142**, and in turn the third guide hole **123**, to the proximal end **112** of the drill guide assembly **100**.

[0032] The orthopedic drill guide assembly **100** further comprises a knob **160** provided at the proximal end **112** of the elongated body. The knob **160** is communicatively connected to the distal sliding portion **142** by the second housing **140** whereby the sliding motion of the distal sliding portion **142** for changing the distance between the third guide hole **123** and the first guide hole **121** is actuated and controlled by turning the knob **160**. By turning the knob **160** in one direction, the second housing **140** along with its sliding portion **142** can be made to slide toward the proximal end **112** decreasing the distance between the first and third guide holes **121**, **123**. By turning the knob **160** in the opposite direction, the second housing **140** and the distal sliding portion **142** can be made to slide away from the proximal end **112** increasing the distance between the first and third guide holes **121**, **123**.

[0033] During use, the drill guide assembly **100** is positioned over a bone repair site (e.g., joint/osteotomy/fracture) so that the first and second guide holes **121**, **122** span over the repair site. In this position, the first guide hole **121** will be on one side of the repair site and the second and the third guide holes **122**, **123** will be on the opposite side of the repair site. Then, a K-wire or a fixation pin (**P1**, **P2**) is placed through each of the first and third guide holes **121** and **123** then driven into the respective bone pieces on the two sides of the repair site. In the illustrated example shown in FIG. 2, one of the holes for the staple is first predrilled into one of the two bone pieces using the first guide hole **121** as a drill guide. Next, a first pin **P1** is inserted through the first guide hole **121** and into the predrilled hole in the first bone piece. Next, a second pin **P2** is inserted through the third guide hole **123** and into the second bone piece. Then, the knob **160** is turned to reduce the distance between the first and third guide holes **121**, **123** thereby compressing the repair site (joint/osteotomy/fracture) until the two bone pieces involved are in contact. After the desired compression is achieved, the repair site should be positioned between the first and the second guide holes **121** and **122**, so that the first guide hole **121** affixed to a first bone piece by the first pin **P1** and the second guide hole **122** is positioned over a second bone piece. Then, the second guide hole **122** is used as a drill guide and a hole is drilled into the second bone piece. Next, the second pin **P2** is removed and the drill guide **100** is removed. The surgeon can now manually compress the repair site and implant a staple into the predrilled holes. The use of the drill guide assembly **100** can be useful for predrilling holes for staples when the joint, osteotomy, or

fracture have some gap between the two bone pieces. In those situations, if a shape memory metal staple is inserted without proper temporary fixation or compression of the two bone pieces, the staple will first bring the two bone pieces together and close the gap as the staple returns to its memorized shape. However, that also brings the legs of the staple inward reducing the amount of compression force the staple exerts on the repair site.

[0034] In some embodiments of the orthopedic drill guide assembly **100**, rather than the knob **160**, a lever can be provided at the proximal end **112** of the elongated body for actuating to pull in the third guide hole **123** toward the first guide hole **121** and reduce the distance between the third guide hole **123** and the first guide hole **121**. The lever would be communicatively connected to the distal sliding portion **142** by a connecting piece that sits within the elongated body **110**, whereby the sliding motion of the distal sliding portion **142** is actuated and controlled by moving the lever.

[0035] In some embodiments, the drill guide assembly **100** is configured so that the distance between the first and second guide holes **121**, **122** is adjustable so that the drill guide assembly **100** can be universally used for various different staple sizes. FIGS. 2B-D illustrate in more detail an example of the internal structure of the drill guide assembly **100** that enables the feature of adjusting the distance between the first and second guide holes **121**, **122**. FIG. 2B is a top view of the drill guide assembly **100**. FIG. 2C shows a longitudinal cross-sectional view of the drill guide assembly **100** taken through the section line B-B shown in FIG. 2B.

[0036] As shown in FIG. 2D, which is an exploded view of the drill guide assembly **100**, the elongated body portion comprises three primary components: the first housing **110**; the second housing **140**, and a core stem **130**. The second guide hole **122** is provided near the distal end **111** of the neck portion **110B** of the first housing **110**.

[0037] The second housing **140** comprises the distal sliding portion **142** near the distal end **111** as discussed above. The second housing **140** is formed of two elongated portions **140A**, **140B**. The two elongated portions are connected at the distal sliding portion **142** but are spaced apart and form a space **140C** therebetween. At the proximal end **112**, the two elongated portions **140A**, **140B** are provided with a threaded portion **144** for threadably engaging the knob **160**.

[0038] In the fully assembled drill guide assembly **100**, the neck portion **110B** of the first housing **110** is received in the space **140C** between the two elongated portions **140A**, **140B**. Thus, near the distal end **111** of the drill guide assembly, the distal sliding portion **142** wraps around the neck portion **110B** as shown in FIGS. 1A and 2A. The proximal half of the two elongated portions **140A**, **140B**, however, are positioned inside the handle portion **110A** of the first housing **110**. To accomplish this arrangement, the handle portion **110A** is only connected to the top surface portion of the neck portion **110B** and forms a spacing **110C** between the handle portion **110A** and the neck portion **110B** as noted in FIG. 2D. The spacing **110C** is open at the bottom side so that the two elongated portions **140A**, **140B** can slip into the spacing **110C** from the bottom side. In this arrangement, the knob **160** rotatably engages the groove **110D** provided at the proximal end **112** of the first housing **110** via the knob's rim **161**. The threaded portion **144** of the second housing **140** is received into the interior **162** of the knob which is threaded with corresponding female threads. This

arrangement can be better seen in the cross-sectional view in FIG. 2C. Thus, by turning the knob 160, the threaded engagement enables moving the second housing 140 and the distal sliding portion 142 back and forth in distal-proximal direction described above.

[0039] The neck portion 110B of the first housing 110 is open from the bottom side noted by the arrow E in FIG. 2D. In the fully assembled drill guide assembly 100, the core stem 130 is slipped into the open space within the neck portion 110B. The neck portion 110B is provided with an opening 116 to accommodate the fixation pin P1 or a drill bit that will be inserted into the first guide hole 121. The opening 116 is elongated to accommodate the full sliding motion of the core stem 130 for adjusting the spacing between the first guide hole 121 and the second guide hole 122. The core stem 130 is provided with a slider button 132 that is positioned between the first guide hole 121 and the proximal end 112 of the drill guide assembly 100.

[0040] The core stem 130 is also provided with a visual marker tab 135. The handle portion 110A can comprise an elongated opening 115 through which the slider button 132 protrudes to allow the user to push pull on the slider button 132 to move the core stem 130 back and forth to move the first guide hole 121 back and forth along a direction (distal-proximal) that is in-line with the longitudinal vertical plane Pv of the drill guide assembly 100.

[0041] The handle portion 110A can also comprise a plurality of slots 114 through which the visual marker tab 135 can at least partially protrude for indicating the position of the first guide hole 121 with respect to the second guide hole 122. Because the first and second guide holes 121, 122 are the drill guide holes that will be used to drill holes for the bone staples, this ability to change the spacing between the first and second guide holes 121, 122 allow the drill guide assembly 100 to be used for many different sizes of staples.

[0042] Therefore, the plurality of slots 114 can be located at preset distances representing particular staple size that corresponds to the selected distance between the first and second guide holes 121, 122. Each of the plurality of slots 114 corresponds to a particular staple size and the visual marker tab 135 can be a colored marking that moves along with the slider button 132 so that the user can see the marker through the slots 114 and tell when a desired spacing between the first and second guide holes 121, 122 has been selected corresponding to the desired staple size. In some embodiments, by having the visual marker tab 135 at least partially protrude through the slots 114, the visual marker tab 135 can be made to click through the slots 114 thus providing additional tactile feedback to the user. As shown in FIG. 2D, the portion 136 of the core stem 130 on which the slider button 132 and the visual marker tab 135 reside can be configured to be a cantilevered structure to allow the slider button 132 and the visual marker tab 135 to be pressed down during the sliding motion. Additionally, when the visual marker tab 135 pops into a slot 114, the visual marker tab can prevent unintended movement of the slider button 132 and lock drill guide assembly to the selected staple size.

[0043] In the drill guide assembly 100, the second guide hole 122 is provided on the first housing 110 at a fixed location. The second guide hole 122 is referred to as being at a fixed location because it is the first guide hole 121 that moves with respect to the position of the second guide hole 122 in the frame of reference of the drill guide assembly 100.

By moving the first guide hole 121, the distance between the first guide hole 121 and the second guide hole 122 can be adjusted.

[0044] The handle portion 110A is shaped to make it comfortable for the user to grab and hold the drill guide 100. The slider button 132 can be provided in the handle portion so that the user can manipulate the slider button 132 with the user's thumb for one-handed operation.

[0045] FIG. 1B illustrates two different top-down views of the slots 114, the slider button 132, and the visual marker tab 135, each view illustrating two different staple size selection. In the view shown on the left side, the slider button 132 is in its most distal position (closest to the distal end 111 of the drill guide assembly. In this example, when the slider button 132 is in this position, the first guide hole 121 is at its closest to the second guide hole 122. Thus, this position represents selection for the smallest staple size. In the view shown on the right side of FIG. 1B, the slider button 132 is in its most proximal position (closest to the proximal end 112 of the drill guide assembly. In this example, when the slider button 132 is in this position, the first guide hole 121 is at its furthest from the second guide hole 122. Thus, this position represents selection for the largest staple size.

[0046] Referring to FIGS. 3A-3B, according to another aspect, a simpler drill guide assembly 100A is disclosed. The drill guide assembly 200 is similar in construction as the drill guide assembly 100A but the drill guide assembly 200 is not configured to perform the pre-drilling temporary compression of the repair site. The drill guide assembly 100A does not have the third guide hole 123 of the drill guide assembly 100. The drill guide assembly 100A comprises the same first and second guide holes 121, 122, the slider button 132 and the plurality of slots 114 as in the drill guide assembly 100. Thus, drill guide assembly 100A is useful for predrilling holes in two bone pieces in a repair site for a staple.

[0047] Referring to FIGS. 4A-4H, an orthopedic drill guide assembly 300 according to another embodiment is disclosed. The drill guide assembly 300 comprises an elongated body 305 having a distal end 311 and a proximal end 312, and a longitudinal vertical plane Pv defined there-through. The elongated body 305 comprises an outer housing 310 having a proximal end and a distal end, and a connecting piece 370 provided within the outer housing 310. The drill guide assembly 300 also comprises a first arm 351 and a second arm 352 retractably connected to the distal end 311 of the outer housing 310. The first arm 351 comprises a distal end 351A and a proximal end 351B. The second arm 352 comprises a distal end 352A and a proximal end 352B.

[0048] The distal ends 351A, 352A of the first and second arms 352, 352, respectively, are provided with a first drill guide sleeve 321 and a second drill guide sleeve 322, respectively. The first drill guide sleeve 321 and the second drill guide sleeve 322 are separated by a spacing D (See FIG. 4B).

[0049] The proximal ends 351B, 352B of the arms are retractably connected to the distal end 311 of the outer housing 310 and configured to move in a retractable motion that changes the spacing D between the first drill guide sleeve 321 and the second drill guide sleeve 322.

[0050] The drill guide assembly 300 further comprises a knob 350 provided at the proximal end 312 of the outer housing 310 for actuating and controlling the retractable

motion of the first and second arms 351, 352. The knob 350 is communicatively connected to the first and second arms 351, 352.

[0051] FIG. 4C shows a longitudinal cross-section of the drill guide assembly 300 where the section is taken through the line A-A shown in FIG. 4B. The connecting piece 370 has a proximal end 372 and a distal end 371 and extends through the outer housing 310 and provides the communicating connection between the knob 350 and the first and second arms 351, 352.

[0052] The knob 350 is connected to the proximal end 372 of the connecting piece 370 and the connection between the knob 350 and the connecting piece 370 is configured to move the connecting piece 370 back and forth between the distal end 311 and the proximal end 312 of the outer housing 310.

[0053] The proximal ends 351B, 352B of the first and second arms 351, 352, respectively, are pivotally connected to a pivot point 380 located at the distal end 371 of the connecting piece 370. The pivot point 380 is on the longitudinal vertical plane Pv. The pivotal connection at the pivot point 380 is made by a pivot pin 381.

[0054] Referring to FIG. 4F, each of the first and second arms 351, 352 are configured with an elongated slot 355, 356, respectively, that extends between the arm's proximal end 351B, 352B and the distal end 351A, 352A. The first and second arms 351, 352 are arranged so that the elongated slot 355, 356 on each of the first and second arms 351, 352 is oriented outward at an acute angle a from the longitudinal vertical plane Pv of the elongated body 305. The first and second arms 351, 352 overlap one another at their proximal ends 351B, 352B where the pivotal connection is made and a portion of the elongated slots 355, 356 overlap one another. The distal end 311 of the outer housing 310 comprises a fixed pin 315 that extends through both of the elongated slots 355, 356 at the overlapping portion, wherein the fixed pin 315 is on the longitudinal vertical plane Pv. As noted above, the pivot point 380 and the fixed pin 315 are both on the longitudinal vertical plane Pv. The fixed pin 315 keeps the two arms 351, 352 from freely pivoting about the pivot point 380 without control. This arrangement allows the user to control the retractable motion of the first and second arms 351, 352 by moving the connecting piece 370 back and forth using the knob 350. The back and forth motion of the connecting piece 370 produces the retractable motion of the first and second arms 351, 352 shown by the arrows 390 in FIG. 4B. The retractable motion of the first and second arms 351, 352 changes the spacing D between the first drill guide sleeve 321 and the second drill guide sleeve 322.

[0055] FIGS. 4B and 4D are illustrations of the drill guide assembly 300 with its first and second arms 351, 352 in their fully extended configuration. The connecting piece 370 has been pushed toward the distal end 311 of the outer housing 310 as far as it can go. In turn, the connecting piece 370 has pushed the first and second arms 351, 352 to their fully extended position because the first and second arms 351, 352 are connected to the connecting piece 370 by the pivot pin 381. As can be seen, each arm has been pushed out to its maximum extension and the fixed pin 315 is at the most proximal ends of the two slots 355, 356. In this configuration, the spacing D between the first and second arms is at its maximum.

[0056] In some embodiments of the drill guide 300, the knob 350 and the connecting piece 370 are connected by

screw thread arrangement that enables the back and forth movement of the connecting piece 370 by turning the knob 350. In the illustrated example shown in FIG. 4C and 4G, the proximal end 372 of the connecting piece 370 is configured with a male screw thread 374. As shown in FIGS. 4C and 4H, the internal bore of the knob 350 is provided with a female screw thread 354 that threadably engage the proximal end 372 of the connecting piece 370.

[0057] In some embodiments of the drill guide 300, the knob 350 can be configured to have a clicking feature as the knob is rotated. Such clicking feature provides both tactile and aural feed back to the user so that the user knows where each position is when the knob is turned to the preset locations correlating to the sizes of the staples. The knob 350 can be provide with grooves 350A as shown in FIG. 4H that interact with a protrusion 350B (See FIG. 4C) on the proximal portion of the outer housing 310.

[0058] In some embodiments of the drill guide 300, each of the first and second drill guide sleeves 321, 322 are provided with a hole sized to receive a drill bit. As shown in FIGS. 4A and 4E, in some embodiments of the drill guide, each of the first and second drill guide sleeves 321, 322 extend parallel to each other.

[0059] Preferably, the connecting piece 370 moves back and forth along the longitudinal vertical plane Pv.

[0060] Referring to FIGS. 5A-5E, according to another aspect of the present disclosure, a surgical bone staple 1000 having a chamfered bridge portion is disclosed. The surgical bone staple 1000 comprises a first leg 1011, a second leg 1012, and a bridge 1020 connecting the two legs. The bridge 1020 has a top surface 1022 along the length of the bridge between a first end and a second end. The top surface 1022 comprises two chamfered edge surfaces 1023 and 1025 that extend substantially the length of the bridge 1020 and one flat surface 1024 between the two chamfered edge surfaces 1023, 1025. The flat surface 1024 also extends substantially the length of the bridge 1020. As shown in the end-view of the staple 1000 in FIG. 5C, the bridge 1020 portion is wider than the width of the first and second legs 1011, 1012. This makes the bridge 1020 stronger and more resistant to unwanted bending in the lateral plane. In this context the lateral plane refers to the plane that is substantially orthogonal to the legs.

[0061] In some embodiments, each of the two legs 1011, 1012 comprises an inner surface 1017 and a plurality of barbs or teeth 1015 configured to prevent backing out of the staple after installation in a bone. Each of the plurality of barbs 1015 are formed by removing material between each barb such that the pointed tips of the barbs do not protrude beyond the inner surface of the legs. This is shown in FIGS. 5B and 5E by the dotted line S that represents the inner surface of the legs.

[0062] Each of the two legs 1011, 1012 comprises a proximal end PE and a distal end DE and each distal end includes a wedge-shaped tip. In this context, the term "proximal" refers to the end of a staple leg that is attached to the bridge 1020 and the term "distal" refers to the end of a staple leg that is away from the bridge 1020. Each of the two ends of the bridge transitions down to a leg. The transition from the bridge 1020 to the legs along the top surface 1024 of the bridge has a first radius of curvature R1 and the transition from the bridge to the legs along the bottom surface of the bridge has a second radius of curvature R2, wherein $R1 > R2$. Preferably, $R1 = 0.110$ inches and $R2 = 0$.

100 inches. In some embodiments, the bone staple **1000** is comprised of a shape memory material such that the staple is movable between an insertion shape and a relaxed shape. **[0063]** FIGS. **5D** and **5E** are illustrations of a bone staple **1000A** that has all the features of the bone staple **1000** discussed above but is a smaller sized staple. According to another embodiment, the bone staple **1000** can have more than two legs. FIG. **5F** shows a bone staple **1000B** that is an example of such variation. The bone staple **1000B** comprises four legs **1011**, **1012**, **1013**, **1014**. Two pairs of legs **1011**, **1013** and **1012**, **1014** are provided. The first pair of legs **1011**, **1013** is on one side of the staple **1000B** and the second pair of legs **1012**, **1014** is on the other side of the staple **1000B**. In some embodiments, each of the legs can have different lengths as desired.

[0064] Referring to FIGS. **10A-10D**, according to another aspect, a bone staple **2000** having an offset bridge portion is disclosed. The bone staple **2000** comprises a pair of legs **2011**, **2012**, each having a proximal end PE and a distal end DE. The pair of legs **2011**, **2012** define a compression plane CP. The bone staple **2000** also comprises a non-linearly extending (i.e., offset) bridge portion **2020** connecting the proximal ends of the pair of legs. The bridge portion **2020** has a generally U-shape and comprises a flat portion **2024** that spans the distance between the proximal ends of the pair of legs. Two ends of the U-shape connect the flat portion **2024** to the proximal ends PE of the legs **2011**, **2012**. The two ends of the U-shape extend laterally between the flat portion **2024** and the proximal ends PE of the legs **2011**, **2012**. The flat portion **2024** is offset from the compression plane CP. The flat portion **2024** has a width that is greater than its thickness to provide enhanced lateral stiffness. The ratio of the width/thickness of the flat portion **2024** can be about 2.0 to 2.5.

[0065] In some embodiments of the bone staple **2000**, the legs **2011**, **2012** comprise inner surfaces **2014** that are facing one another and a plurality of barbs **2015** along the inner surfaces **2014**. In some embodiments, the bone staple **2000** is comprised of a shape memory material such that the staple is movable between an insertion shape and an implanted shape.

[0066] FIG. **10C** is a side view of the bone staple **2000**. FIG. **10D** is an illustration showing an example of three different size offset bridge bone staples in a nested arrangement. This illustration of the nested arrangement shows that the bone staple embodiment **2000** of different sizes can be used to place multiple staples in close proximity to each other in the patient.

[0067] Referring to FIGS. **12A-12E**, a bone staple **3000** according to another embodiment is disclosed. Two or more pairs of legs **3011**, **3012**, **3013**, **3014** extend from the bridge **3020** all in the same direction. Each of the legs **3011**, **3012**, **3013**, **3014** has a surface that faces the other leg in its pair, where the surface comprises a plurality of barbs **3015**. In the illustrated example, the bone staple **3000** is shown with two pairs of legs: **3011** and **3012** being one pair and **3013** and **3014** being the other pair. As shown, each pair of legs have a plurality of barbs **3015** on the interior surface that faces the other leg in the pair. The bone staple **3000** is comprised of a shape memory material such that the staple is movable between an insertion shape (shown in FIG. **12B**) and a relaxed shape (shown in FIG. **12D**).

[0068] The bone staple **3000** is not an intermedullary device and the bridge **3020** has an open-loop configuration

and has a thickness T of about 0.05-0.1 inches. This thickness of the bridge **3020** allows easy removal of the staple from the repair site because it is easy to grab.

[0069] The configuration of the staple **3000** shown in FIG. **12A** is its as-fabricated shape. For example, the staple **3000** can be machined out of a shape memory metal tubing (e.g. Nitinol) and retain the circular cross-sectional shape of the tube. FIG. **12C** is the top down view showing the circular outline of the open-loop bridge **3020** as-fabricated from a tube.

[0070] FIG. **12D** shows the outline of the open-loop bridge **3020** that has been compressed into an elongated oval-like shape and heat set to set the memory metal's memorized shape. The locations of the legs **3011**, **3012**, **3013**, and **3014** are shown by the dotted lines. The spacing between the legs of each pair of legs is set to d1.

[0071] FIGS. **12B** and **12E** show the insertion shape of the open-loop bridge **3020** after it has been cold worked into a shape to be held by a staple holder/insertor. The configuration shown in FIGS. **12B** and **12E** is achieved by bending the two sections of the bridge **3020**, one section between the first pair of legs **3011** and **3012**, and another section between the second pair of legs **3013** and **3014**, straight. The result is that the legs in each pair are moved further apart to a larger spacing d2. This configuration is the shape of the bone staple **3000** to be held by an inserting tool for implantation. Four holes are first drilled into the bone repair site using a specific drill guide for that purpose. Then, the four legs of the bone staple **3000** in the insertion shape shown in FIGS. **12B**, **12E** is inserted into those holes. As the bone staple **3000** reaches the activation temperature in the patient's body, the bridge **3020** of the bone staple **3000** returns to the memory set shape (i.e., the relaxed shape) shown in FIG. **12D**. During this transformation, the two straight segments of the staple **3000** will move outward as indicated by the arrows E in FIG. **12E**. This causes the other two segments of the bridge **3020** (the segments on top and bottom of FIG. **12E** to move inward toward each other as indicated by the arrows C. This will generate compression force in the same direction as the arrows C as the legs try to return to the relaxed shape of FIG. **12D**.

[0072] Referring to FIGS. **13A-13D**, a bone staple **4000** having four staple legs in offset arrangements according to some other embodiments are disclosed. In these figures, the bone staple **4000** is shown in its insertion shape, as the staple legs are all shown in parallel arrangement. The bone staple **4000** comprises a bridge **4020** having an elongated shape with a longitudinal axis L defined through its length. The bridge **4020** further comprises a first corner **4021**, a second corner **4022**, a third corner **4023**, and a fourth corner **4024**. The bone staple **4000** also comprises a first leg **4011** extending from the first corner **4021**, a second leg **4012** extending from the second corner **4022**, a third leg **4013** extending from the third corner **4023**, and a fourth leg **4014** extending from the fourth corner **4024**. As shown, the first leg **4011** and the third leg **4013** are configured to be the pair providing the necessary compressive force when the staple is implanted into a bone repair site. The second leg **4012** and the fourth leg **4014** are configured to be the second pair providing the necessary compressive force when the staple is implanted into a bone repair site. As shown in FIG. **13A**, the two legs that are paired for providing the compression function have a plurality of barbs **4015** on the surface that

face each other. The barbs **4015** are oriented to prevent the staple from backing out once the staple is placed into a bone.

[0073] The bridge **4020** can also comprise a central reinforcement ridge **4027** that extends down the length of the bridge **4020**. The central ridge **4027** is thicker than the rest of the bridge and reinforces the strength and rigidity of the bridge **4020**. When the thickness of the central ridge **4027** is appropriately thick, the rigidity of the bridge **4020** can be comparable to a bone plate. In some embodiments, the central ridge **4027** is the thickest near the center of the bridge **4020** and tapers to the flat portion of the bridge **4020** near the two ends of the bridge **4020** to provide a smooth profile.

[0074] The bridge **4020** comprises a first section that includes the first and second corners **4021**, **4022** and a second section that includes the third and fourth corners **4023**, **4024**. The first and second corners **4021**, **4022** are arranged such that a line **L1** connecting the first and second corners **4021**, **4022** intersect the longitudinal axis **L** of the bridge **4020** at a first non-orthogonal angle θ_1 .

[0075] In some embodiments of the bone staple **4000**, the third and fourth corners **4023**, **4024** are arranged such that a line **L2** connecting the third and fourth corners **4023**, **4024** intersect the longitudinal axis of the bridge at a second non-orthogonal angle θ_2 . In some embodiments, as shown in FIG. 13A, the first non-orthogonal angle θ_1 and the second non-orthogonal angle θ_2 are different. Thus, the bone staple **4000** is configured so that the first pair of staple legs **4011**, **4013** have greater spacing between them compared to the second pair of staple legs **4012**, **4014**. The first pair of staple legs **4011**, **4013** can span across a greater distance in the bone compared to the second pair of staple legs **4012**, **4014**. An example of use for such bone staple **4000** is for metatarsophalangeal (MTP) fusion. Having four offset legs, the bone staple **4000** can be much more robust in being useful in various situations depending on the particular arrangement and physiology of the patient. The bone staple **4000** can be fabricated to have different offset configuration to fit the needs of different situation.

[0076] In some embodiments, the first non-orthogonal angle θ_1 and the second non-orthogonal angle θ_2 can be the same angle. An example of such configuration is shown in the bone staple **4000A** shown in FIG. 13D. In this embodiment, the first pair of staple legs **4011A**, **4013A** and the second pair of staple legs **4012A**, **4014A** have the same spacing between the legs in each pair. The two pairs are just offset from each other and parallel to the longitudinal axis **L** of the bridge portion **4020**. Thus, the four corners **4021A**, **4022A**, **4023A**, **4024A** and their respective staple legs **4011A**, **4012A**, **4013A**, **4014A** each form a parallelogram arrangement as shown in FIG. 13D.

[0077] As shown in the embodiment of the bone staple in FIG. 13D, the line **L1** connecting the first and second corners **4021A**, **4022A** and the line **L2** connecting the third and fourth corners **4023A**, **4024A** can be parallel.

[0078] According to some embodiments, each of the four legs on the staple **4000**, **4000A** can have a length that is independent of one another as can be seen in FIG. 13B to accommodate particular anatomical locations. For example, the bone staple **4000** shown in FIG. 13B where the two legs **4011** and **4012**, which are one leg in each of the two pairs, are longer than their paired legs **4013** and **4014**, respectively, can be useful where one side of the bone repair/fusion site has a larger bone piece than the other. The longer legs would go into the larger bone piece.

[0079] In some embodiments, all four of the staple legs on the bone staple **4000**, **4000A** can be of the same length.

[0080] In some embodiments, the bridge **4020** has an upper surface that curves down ward from the center line **L** to the four corners **4021**, **4022**, **4023**, **4024**.

[0081] According to an aspect of the present disclosure, FIGS. 6A-7D show a bone staple insertion system. The bone staple insertion system comprises a staple holding cartridge **200** and a universal staple inserter **400**. FIGS. 6A-6B show a cartridge **200** configured for holding a two-legged inline bone staple such as the staple **1000** (shown in FIG. 5A) that facilitates staple insertion procedure.

[0082] The cartridge **200** comprises an outer frame **210** and an inner frame **220**. The inner frame **220** is configured to releasably hold a bone staple **1000** in its insertion shape. In the insertion shape, the bridge portion **1020** of the staple **1000** is straightened with the two legs **1011** and **1012** opened so that they are substantially parallel to each other.

[0083] The inner frame **220** comprises a base portion **221** and a pair of restraining arms **222A** and **222B** that extend upward from the base portion **221**. Each of the restraining arms **222A**, **222B** has a restraining tab **223A**, **223B**, respectively, at the top end of the restraining arms **222A**, **222B**. The restraining tabs **223A**, **223B** extend inward toward each other and meet in the middle of the space between the two restraining arms **222A**, **222B**.

[0084] As shown in FIGS. 6A and 6B, the bone staple **1000** is loaded onto the inner frame **220** so that the two legs **1011**, **1012** of the staple **1000** are held in the open configuration (i.e., the two legs substantially parallel to each other), preventing them from closing back to their relaxed position, and the bridge portion **1020** is braced against the restraining tabs **223A**, **223B**.

[0085] To help keep the two legs **1011**, **1012** of the bone staple **1000** in the open configuration, the base portion **221** of the inner frame **220** is provided with two recesses **221A** and **221B**, into which the two staple legs are inserted. This can be better seen in FIG. 6A.

[0086] The two restraining tabs **223A**, **223B** are not permanently connected to each other so the staple **1000** can be removed from the inner frame **220** by spreading the two restraining arms **222A** and **222B** apart.

[0087] The base portion **210** of the cartridge **200** comprises two guiding towers **211**, **212** that are spaced apart at a set distance for receiving and guiding the universal staple inserter **400**. FIGS. 7A-7D show the engaging action of the cartridge **200** and the universal staple inserter **400**. The universal staple inserter **400** is configured so that the staple inserter **400** and the configuration of the cartridge **200** enables transfer of the bone staple **1000** from the cartridge **200** to the universal staple inserter **400** in one single engaging motion. The engaging motion is the simple motion of inserting the universal staple inserter **400** between the two guiding towers **211**, **212** straight down as represented by the downward arrow in FIG. 7A.

[0088] The universal staple inserter **400** comprises two pairs of staple grabbers **410A**, **410B**. Each pair of staple grabbers **410A**, **410B** has two arms, one on each side of the inserter **400**, extending downward. The leading ends of the staple grabbers **410A**, **410B** have retaining tabs **411A**, **411B** that extend downward beyond the body of the universal staple inserter **400**.

[0089] As the universal staple inserter **400** is lowered into the cartridge assembly **200** between the two guiding towers

211, 212, the retaining tabs 411A, 411B of the staple grabbers 410A, 410B encounter the bridge portion 1020 of the bone staple 1000 causing the retaining tabs 411A, 411B, and in turn the staple grabbers 410A, 410B, to spread apart. The retaining tabs 411A, 411B have a wedge-like shape as shown in FIG. 7C to facilitate this action. As the universal staple inserter 400 is pushed down further, the retaining tabs 411A, 411B slides over the bridge portion 1020 then snaps back toward their resting position so that the arms of each pair of the staple grabbers 410A, 410B close together from the opposite sides with the bridge portion 1020 of the staple 1000 caught in between. At this stage, the retaining tabs 411A, 411B are now underneath the bridge portion 1020. This position is shown in FIG. 7C. The wedge-like shape of the retaining tabs 411A, 411B helps to retain the staple 1000.

[0090] Additionally, as the universal staple inserter 400 is being lowered into the cartridge assembly 200 and the retaining tabs 411A, 411B are passing by the bridge portion 1020, the leading end of the main body of the universal staple inserter 400 gets wedged in between the two restraining tabs 223A, 223B of the inner frame 220 and urge them apart so that the restraining tabs 223A, 223B no longer are holding down the staple as the retaining tabs 411A, 411B of the staple grabbers 410A, 410B grab hold of the staple 1000.

[0091] With the transfer of the staple 1000 from the inner frame 210 of the cartridge 200 to the universal staple inserter 400 complete, the inserter 400 can now be removed from the cartridge assembly 200 as shown in FIG. 7D. The universal staple inserter 400 is now holding the staple 1000. The retaining tabs 411A, 411B of the staple grabbers 410A, 410B keep the bridge portion 1020 of the staple 1000, which is normally bowed upward in the relaxed shape as shown in FIG. 5A, substantially straight as shown. This maintains the staple 1000 in its insertion shape with its two legs in substantially parallel configuration so that the staple can be inserted into a bone repair site. The universal staple inserter 400 is configured so that the two pairs of staple grabbers 410A, 410B can be quickly spread apart and release the staple after the staple has been inserted into a bone repair site.

[0092] The cartridge assembly 200 and the associated universal staple inserter 400 allows one staple inserter to be universally used with staples of a variety of sizes. As long as the central portions of the bridge portion 1020 of the staples of varying sizes are configured to have the same bowed shape, the universal staple inserter 400 can be used in conjunction with the cartridge assembly 200 holding each of the staples. The cartridge assembly 200 would be provided in different sizes to hold different size staples. The two guiding towers 211, 212 of the cartridge assemblies 200 will all be the same size to receive the universal staple holder 400.

[0093] FIG. 8 is an illustration of an overlapping view of three different size staples 1000, 1000C, and 1000D that are configured for using with the universal staple inserter 400. The bridge portions of the three staples have a Central Portion that have the same bowed shape.

[0094] Referring to FIGS. 14A-14I, a bone staple inserter system according to another embodiment is disclosed. The bone staple inserter system comprises a cartridge 777 shown in FIG. 14A. A bone staple such as the two-legged inline staple 1000 shown in FIG. 5A is securely mounted to the cartridge 777. As described above, the bone staple 1000 comprises a first leg and a second leg oriented toward each

other in a relaxed state, and a curved bridge portion 1020 connecting the first and second legs, where the cartridge 777 comprises a first end and a second end, the first end provided with a channel 710 that is sized to receive the bridge portion 1020 of the bone staple 1000 and securely hold the bone staple 1000 by an interference fit.

[0095] The bone staple inserter system also comprises a bone staple inserter 700A that is configured to receive the cartridge 777.

[0096] Referring to FIG. 14A, the cartridge 777 also comprises a connector part 720 that is configured to securely attach to the bone staple inserter 700A. In the illustrated example, the connector part 720 is a cylindrical structure and comprises two tabs 722 configured for releasably attaching to the bone staple inserter 700A by a twisting motion.

[0097] Referring to FIGS. 14B-14D, the bone staple inserter 700A has a body 750. The body has a cartridge engaging end 751 that is configured with a recess 720A for receiving the connector part 720 of the cartridge. Around the periphery of the recess 720A is provided with two L-shaped cutouts 722A for receiving the two tabs 722 on the connector part 720 of the cartridge 777. The L-shape of the cutouts 722A allows the tabs 722 to first engage the cartridge engaging end 751 straight then twisted to temporarily lock the cartridge 777 with the inserter 700A.

[0098] The inserter 700A further comprises a pushrod 770 that protrudes through the recess 720A for engaging the cartridge 777 and release the staple 1000 from the cartridge 777. The pushrod 770 is configured to move in and out of the cartridge engaging end 751 as indicated by the arrow A in FIG. 14B. The movement of the pushrod 770 is controlled by a lever 760. The lever 760 is connected to the pushrod 770 on the opposite end of the pushrod 770 by a hinged connection so that when the lever 760 is pulled up as shown in FIG. 14B, the pushrod 770 is in its retracted position. When the lever 760 is pushed down into the body 750 as shown in FIG. 14C, the pushrod 770 is in its fully extended position.

[0099] An exemplary mechanical arrangement that enables the reciprocating motion of the pushrod 770 by the manipulation of the lever 760 is shown in the cross-sectional view of the staple inserter 700A shown in FIG. 14G. In this example, the pushrod 770 is connected to the lever 760 via an intermediate linkage 770A. The connections between the intermediate linkage 770A and the pushrod 770 and the lever 760 are both hinged connections to accommodate the angular bending at those connection points when the lever 760 goes from the pushed-down position of FIG. 14G to the pulled-up position of FIG. 14B. The lever 760 is connected to the body 750 by a hinge pin 765 as shown in the side view of FIG. 14H. The hinge pin 765 defines the fulcrum F (shown in FIG. 14G) and provides the hinged movement for the lever 760. The hinged connection between the intermediate linkage 770A and the lever 760 is identified by the point H in FIG. 14G. Because the hinge point H is between the pushrod 770 and the fulcrum F, when the lever 760 is pulled up, the hinge point H rises with the lever 760 and the intermediate linkage 770A pulls the pushrod 770 into the body 750 into its retracted position. When the lever 760 is pushed back down, the reverse happens and the intermediate linkage 770A pushes the pushrod 770 out to its extended position.

[0100] Referring to FIG. 14E which shows the empty cartridge 700B attached to the inserter 700A, the pushrod

770 in its fully extended position can be seen. In this position, the pushrod protrudes into the channel 710. Thus, when a bone staple 1000 is securely loaded into the channel 710 by a press-fitting or an interference-fitting, the bowed portion of the staple's bridge 1020 sits within the channel 710. In use, the lever 760 on the staple inserter 700A is first pulled up to full retract the pushrod 770. Then, the connector part 720 of a loaded cartridge 700B is inserted into the recess 720A of the staple inserter 700A and twisted to lock the engagement. At this point, the staple 1000 is in its relaxed shape and the two legs of the bone staple 1000 are biased toward each other as shown in FIG. 14A. Next, the staple 1000 is changed into its insertion shape by pushing the lever 760 down to the position shown in FIGS. 14C and 14E. This extends the pushrod 770 into the channel 710 which pushes on the bridge 1020 of the staple 1000 from the convex side of the bowed bridge 1020. This straightens the bridge 1020 causing the two legs of the staple 1000 to be substantially parallel to each other. With the staple 1000 engaged with the inserter 700A in this manner and the lever 760 is in the down position, the angle of the intermediate linkage 770A creates an over center connection where the staple pushing back on the pushrod 770 keeps the linkage formed by the pushrod 770, the intermediate linkage 770A, and the lever 760 rigid (similar to a person locking a straightened elbow) and keeps the system engaged and prevents the lever from popping up to the pulled up position when not desired. Now, the staple 1000 is in its insertion shape and the surgeon can insert the staple 1000 into the pre-drilled holes in the bone repair site. After the insertion, the inserter 700A/cartridge 700B assembly can be removed by pulling up on the lever 760 to retract the pushrod 770 and relieve the pressure on the bridge 1020 returning the staple 1000 to its relaxed shape. Because the staple 1000 is now held only by the friction between the channel 710 in the cartridge 700B and the bridge 1020, the inserter 700A/cartridge 700B assembly can be removed by laterally twisting the assembly.

[0101] Referring to FIG. 9, according to an aspect of the present disclosure, a bone staple inserter kit 500 for implanting two-legged inline bone staples such as those illustrated in FIGS. 5A-5E is disclosed. The bone staple inserter kit 500 comprises a first sterile package 500A containing a first bone staple securely mounted to a first cartridge assembly 200 described above. The first bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion connecting the first and second legs.

[0102] The bone staple inserter kit 500 also comprises second sterile package 500B containing a bone staple inserter 400 described above that is configured to receive the cartridge assembly 200 loaded with a bone staple. The second sterile package 500B also contains a universal drill guide assembly (like the drill guide assemblies 100, 100A, or 300). The second sterile package 500B can also contain other hardware components 10 necessary to use the drill guide. Some examples are one or more drill bits, fixation pins, etc.

[0103] In some embodiments, the bone staple inserter kit 500 can further comprise one or more additional sterile packages, like the sterile package 500A, each containing an additional cartridge assembly securely holding a bone staple, where the bone staples in the additional sterile packages can be of same or difference size as the bone staple in the first sterile package.

[0104] By providing bone staple inserter kits such as 500 described herein, only one set of universal instruments need to be provided in one sterile package 500B and several different sterile packages 500A each containing a different size staple can be provided.

[0105] Referring to FIG. 15, according to an aspect of the present disclosure, a bone staple inserter kit 800 is disclosed. Such kit comprises a first sterile package 801A containing a cartridge assembly 777 and a second sterile package 802 containing a set of universal instruments needed for implanting bone staples.

[0106] The cartridge assembly 777 provided in the first sterile package 801A comprises a bone staple securely mounted to the first cartridge as described herein with reference to FIG. 14A. The universal instruments provided in the second sterile package 802 can comprise the staple inserter 700A described with reference to FIG. 14B. The second sterile package 802 preferably also comprises one or more of the universal drill guide 100, 100A, and 300 described herein.

[0107] In some embodiments, the bone staple inserter kit 800 can further comprise one or more additional first sterile packages 801B, 801C, etc. Each of these additional first sterile packages can contain a cartridge assembly holding a staple of different size or a same size as the staple in the first sterile package 801A.

[0108] Although the devices, kits, systems, and methods have been described in terms of exemplary embodiments, they are not limited thereto. Rather, the appended claims should be construed broadly, to include other variants and embodiments of the devices, kits, systems, and methods, which may be made by those skilled in the art without departing from the scope and range of equivalents of the devices, kits, systems, and methods.

We claim:

1. A bone staple comprising:

- a first leg having a wedge-shaped tip;
- a second leg having a wedge-shaped tip, wherein each of the first and second legs further includes an inner surface each having a plurality of barbs, each barb having a tip that is located flush with the inner surface of the leg; and
- a bridge extending between a first end of the first leg and a second end of the second leg, the bridge comprising a top surface along the length of the bridge, wherein the top surface comprises two chamfered edge surfaces that extend substantially the length of the bridge and one flat surface located between the two chamfered edge surfaces, wherein the flat surface extends substantially the length of the bridge.

2. The bone staple of claim 1, the bridge forms a first transition to the first leg at the first end and a second transition to the second leg at the second end wherein each of the first end transition and the second end transition has a first radius of curvature R1 and a transitions from the bridge to the legs along a bottom surface of the bridge has a second radius of curvature R2, wherein $R1 > R2$.

3. The bone staple of claim 1, wherein the staple is comprised of a shape memory material such that the staple is movable between an insertion shape and a relaxed shape.

4. A bone staple comprising:

- a pair of spaced-apart legs, each having a proximal end and a distal end, the pair of spaced-apart legs defining a compression plane;

a non-linearly extending bridge connecting the distal ends of the pair of spaced-apart legs, wherein the bridge has a generally U-shape that comprises:
 a flat portion that spans the distance between the distal ends of the pair of spaced-apart legs; and
 two ends of the U-shape connecting the flat portion to the proximal ends of the spaced-apart legs, wherein the two ends of the U-shape extending laterally between the flat portion and the proximal ends of the spaced-apart legs, whereby the flat portion is off-set from the compression plane, wherein the flat portion has a width that is greater than its thickness.

5. The bone staple of claim 4, wherein each of the pair of spaced-apart legs comprises a surface arranged confronting spaced-apart relation to one another each having a plurality of barbs along the surface.

6. The bone staple of claim 5, wherein the staple is comprised of a shape memory material such that the staple is movable between an insertion shape and a relaxed shape.

7. A bone staple inserter kit comprising:

a first sterile package containing a first bone staple securely mounted to a first cartridge, wherein the first bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion connecting the first and second legs, wherein the first cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge portion of the first bone staple and securely hold the first bone staple by an interference fit; and

a second sterile package containing; a bone staple insertion tool that is configured to receive a cartridge, like the first cartridge that has a bone staple, like the first bone staple, securely mounted thereon; a universal drill guide assembly; and a drill bit, wherein the bone staple insertion tool comprises an actuation mechanism that can transform the bone staple to an insertion shape after the cartridge with the bone staple mounted thereon is mounted on the insertion tool.

8. The bone staple inserter kit of claim 7, wherein the actuation mechanism applies pressure on the bridge portion of the staple to straighten the curved bridge portion after the cartridge that has the bone staple is mounted on the insertion tool, wherein straightening the curved bridge portion expands the first and second legs of the bone staple into the insertion shape.

9. The bone staple inserter kit of claim 7, further comprising a third sterile package containing a second bone staple securely mounted to a second cartridge, wherein the second bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion connecting the first and second legs, wherein the second cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge portion of the second bone staple and securely hold the second bone staple by an interference fit, wherein the first bone staple and the second bone staple have difference sizes.

10. A surgical staple comprising:

a bridge having an elongated shape with a longitudinal axis defined through its length, the bridge further comprising first, second, third, and fourth corners;
 a first leg extending from the first corner;
 a second leg extending from the second corner;

a third leg extending from the third corner; and

a fourth leg extending from the fourth corner;

wherein the bridge comprises a first section that includes the first and second corners, and a second section that includes the third and fourth corners; and

wherein the first and second corners are arranged such that a line connecting the first and second corners intersect the longitudinal axis of the bridge at a first non-orthogonal angle.

11. The surgical staple of claim 10, further wherein the third and fourth corners are arranged such that a line connecting the third and fourth corners intersect the longitudinal axis of the bridge at a second non-orthogonal angle.

12. The surgical staple of claim 11, wherein the first non-orthogonal angle and the second non-orthogonal angle are the same angle.

13. The surgical staple of claim 10, wherein the line connecting the first and second corners and the line connecting the third and fourth corners are parallel.

14. The surgical staple of claim 10, wherein the bridge further comprising a central reinforcement ridge that is thicker than the rest of the bridge and extending along the longitudinal axis in a portion of the length of the bridge.

15. The surgical staple of claim 10, wherein each of the legs have a length that is independent of one another.

16. The surgical staple of claim 10, wherein the legs are of the same length.

17. The surgical staple of claim 10, wherein the bridge has an upper surface that curves down ward from the center line to the four corners.

18. A system comprising;

a bone staple including a pair of spaced-apart legs, each having a proximal end and a distal end, the pair of spaced-apart legs defining a compression plane;

a bridge connecting the distal ends of the pair of spaced-apart legs, wherein the bridge has a generally curved shape with a flat portion that spans the distance between the distal ends of the pair of spaced-apart legs and two ends coupling the flat portion to the proximal ends of the spaced-apart legs, wherein the two ends extend laterally between the flat portion and the proximal ends of the spaced-apart legs;

a first sterile package containing the bone staple securely mounted to a first cartridge, wherein the pair of spaced-apart legs are oriented toward each other in a relaxed state, wherein the first cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge of the bone staple and securely hold the bone staple by an interference fit; and

a second sterile package containing a bone staple insertion tool that is configured to receive the cartridge so as to secure the bone staple for implantation in a patient wherein securing the bone staple for implantation by the bone staple insertion tool comprises transforming the bone staple to a biased insertion shape after the cartridge with the bone staple mounted thereon is mounted on the insertion tool.

19. The system for delivering a bone staple to a surgical theater of claim 18, wherein transforming the bone staple to a biased insertion shape includes applying pressure on the bridge of the bone staple to straighten the bridge after the cartridge that has the bone staple is mounted on the insertion

tool, wherein straightening the bridge expands and thereby biases the first and second legs of the bone staple into the insertion shape.

20. The system for delivering a bone staple to a surgical theater of claim **18**, wherein, further comprising a third sterile package containing a second bone staple securely mounted to a second cartridge, wherein the second bone staple comprises a first leg and a second leg oriented toward each other in a relaxed state, and a curved bridge portion connecting the first and second legs, wherein the second cartridge comprises a first end and a second end, the first end provided with a channel that is sized to receive the bridge portion of the second bone staple and securely hold the second bone staple by an interference fit, wherein the first bone staple and the second bone staple have difference sizes.

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